



Linear Algebra - Suresh Sir' notes

Mathematics I (Birla Institute of Technology and Science, Pilani)



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Note: Some concepts of Linear Algebra are briefly described here just to help the students. Therefore, the following study material is expected to be useful but not exhaustive for the Mathematics-II course. For detailed study, the students are advised to attend the lecture/tutorial classes regularly, and consult the text book prescribed in the hand out of the course.

Chapter 2 (2.1-2.4)

Elementary row operations

There are three elementary row operations:

- (1) Interchanging two rows R_i and R_j (symbolically written as $R_i \leftrightarrow R_j$)
- (2) Multiplying a row R_i by a non-zero number k (symbolically written as $R_i \rightarrow kR_i$)
- (3) Adding constant k multiple of a row R_j to a row R_i (symbolically written as $R_i \rightarrow R_i + kR_j$)

To see how row transformations are applied, consider the matrix

$$A = \begin{bmatrix} 4 & 8 & 10 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_2$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 10 \\ 3 & 5 & 6 \end{bmatrix}.$$

Applying $R_2 \leftrightarrow (1/2)R_2$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}.$$

Applying $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$, we get

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \\ 0 & -1 & -3 \end{bmatrix}.$$

Note. The matrices resulting from the row transformation(s) are known as row equivalent matrices. That is why the sign of equivalence $\sim$ is used after applying row transformations. So we can write

$$A = \begin{bmatrix} 4 & 8 & 10 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 10 \\ 3 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \\ 0 & -1 & -3 \end{bmatrix}.$$

Notice that row equivalent matrices can be obtained from each other by applying suitable row transformation(s), but row equivalent matrices need not be equal.

Row Echelon Form

A matrix is said to be in row echelon form¹ (REF) if

- (i) all the zero rows, if any, lie at the bottom, and
- (ii) the leading entry (the first non-zero entry) in any row is 1, and its column lies to the right of the column of the leading entry of the preceding row.

In addition, if all the entries above the leading entries are 0, then the matrix is said to be in reduced row echelon form (RREF).

Ex. $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ all are in REF.

Ex. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$ all are in RREF.

The following example illustrates that how we find REF and RREF of a given matrix.

Ex. Find RREF of the matrix

$$A = \begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix}.$$

Sol. We have

$$A = \begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_2$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}.$$

Applying $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$, we get

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \\ 0 & -1 & -3 \end{bmatrix}.$$

Applying $R_2 \leftrightarrow R_3$, we obtain

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -3 \\ 0 & 0 & -1 \end{bmatrix}.$$

Applying $R_2 \rightarrow -R_2$ and $R_3 \rightarrow -R_3$, we get

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}. \text{ Notice that it is REF of } A.$$

¹**Dictionary meaning of echelon:** A formation of troops in which each unit is positioned successively to the left or right of the rear unit to form an oblique or steplike line.

Applying $R_2 \rightarrow R_2 - 3R_3$ and $R_1 \rightarrow R_1 - 3R_3$, we get

$$A \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Finally, applying $R_1 \rightarrow R_1 - 2R_2$, we get

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

the RREF of A .

Useful Tip: From the above example, one may notice that for getting RREF of a matrix we make use of first row to make zeros in the first column, second row to make zeros in the second column and so on.

Note: REF of matrix is not unique, but RREF is unique.

Inverse of a Matrix

Let A be a matrix with m rows say $R_1, R_2, \dots, R_m$, and B be a matrix with n columns say $C_1, C_2, \dots, C_n$. Then the product matrix AB is of order $m \times n$, and

$$A.B = \begin{bmatrix} R_1 \\ R_2 \\ \dots \\ R_m \end{bmatrix} [C_1 \ C_2 \ \dots \ C_n] = \begin{bmatrix} R_1 C_1 & R_1 C_2 & \dots & R_1 C_n \\ R_2 C_1 & R_2 C_2 & \dots & R_2 C_n \\ \dots & \dots & \dots & \dots \\ R_m C_1 & R_m C_2 & \dots & R_m C_n \end{bmatrix} = AB$$

If we interchange two rows in A say $R_1 \leftrightarrow R_2$, then the first two rows of AB also get interchanged. Similarly, it is easy to see that applying any of the other two row operations in A is equivalent to applying the same row operation in AB . Thus, we conclude that applying any elementary row operation in the matrix A is equivalent to applying the same elementary row operation in the matrix AB . Hence, if R is any row operation, then $R(AB) = R(A)B$. Note that the matrix B is left unchanged. We make use of this fact to find the inverse of a matrix.

Let A be a given non-singular matrix of order $n \times n$. To find the inverse of A , first we write $A = I_n A$. In this identity, we apply the elementary row operations on the left hand side matrix A in such a way that it transforms to I_n , the RREF of A . As discussed above, the same row operations apply to the first matrix I_n on right hand side and suppose it transforms to a matrix B . Then, we have $I_n = BA$. Therefore, $A^{-1} = B$. This method for obtaining the inverse of a matrix is called **Gauss-Jordan method**.

Note: One may use elementary column operations ($C_i \leftrightarrow C_j$, $C_i \rightarrow kC_i$ and $C_i \rightarrow C_i + kC_j$) also to find A^{-1} . In this case, we write $A = AI_n$ and apply elementary column operations to obtain $I_n = AB$ so that $A^{-1} = B$. **It may be noted that we can not apply row and column operations together for finding A^{-1} in the Gauss-Jordan method.**

Ex. Use Gauss-Jordan method to find inverse of the matrix

$$A = \begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix}$$

Sol. We write $A = I_3 A$ and therefore

$$\begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A.$$

Applying $R_1 \leftrightarrow R_2$, we obtain

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A.$$

Applying $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \\ 0 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & -3 & 1 \end{bmatrix} A.$$

Applying $R_2 \leftrightarrow R_3$, we obtain

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -3 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -3 & 1 \\ 1 & -2 & 0 \end{bmatrix} A.$$

Applying $R_2 \rightarrow -R_2$ and $R_3 \rightarrow -R_3$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 3 & -1 \\ -1 & 2 & 0 \end{bmatrix} A.$$

Applying $R_2 \rightarrow R_2 - 3R_3$ and $R_1 \rightarrow R_1 - 3R_3$, we get

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -5 & 0 \\ 3 & -3 & -1 \\ -1 & 2 & 0 \end{bmatrix} A.$$

Finally, applying $R_1 \rightarrow R_1 - 2R_2$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 2 \\ 3 & -3 & -1 \\ -1 & 2 & 0 \end{bmatrix} A.$$

$$\therefore A^{-1} = \begin{bmatrix} -3 & 1 & 2 \\ 3 & -3 & -1 \\ -1 & 2 & 0 \end{bmatrix}$$

Useful Tip: To find inverse of A , first write $A = I_n A$. Then change the left hand side matrix A to its RREF by applying suitable row transformations so that $I_n = BA$ and $A^{-1} = B$.

Note: You might be familiar that inverse of a square matrix A exists if and only if A is non-singular, that is, $|A| \neq 0$. Note that RREF of a non-singular matrix is always a unit matrix.

Note: You know that when a system of n linear equations in n variables is represented in the matrix form $AX = B$, then A is n square matrix of the coefficients of the variables, and solution of the system reads as $X = A^{-1}B$ provided A^{-1} exists. In case, if number of equations is not equal to the number of variables, then A is not a square matrix, and therefore A^{-1} is not defined. In what follows, we present a general strategy for solving a system of linear equations. First we introduce the concept of rank of a matrix.

Rank of a Matrix

Let A be a matrix of order $m \times n$. Then rank of A , denoted by $\text{rank}(A)$, is defined as the number of non-zero rows in the REF of the matrix A .

Ex. Find rank of the matrix $A = \begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix}$.

Sol. Applying suitable row transformations, we obtain

$$A = \begin{bmatrix} 2 & 4 & 5 \\ 1 & 2 & 3 \\ 3 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

We see that the REF of A carries three non-zero rows. So rank of A is 3.

Useful Tip: To find rank of a matrix A , find REF of A . Then rank of A is the number of non-zero rows in REF of A .

Additional Information: The number of non-zero rows in REF of A is, in fact, defined as the row rank of A . Similarly, the number of non-zero rows in REF of A' (transpose of A) is defined as the column rank of A . Further, it can be established that the row rank is always equal to the column rank. It follows that rank of A is less than or equal to the minimum of m (number of rows in A) and n (number of columns in A).

Solution of a System of Linear Equations

A system of m linear equations in n unknown variables $x_1, x_2, \dots, x_n$ is given by

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2, \\ &\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m. \end{aligned}$$

Here a_{ij} and b_i ($i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$) are constants. The matrix form of this system of linear equations reads as

$$AX = B,$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}.$$

The system $AX = B$ is said to be **non-homogeneous** provided $B \neq O$. Further, $AX = O$ is called **homogeneous** system. The matrix

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix},$$

which is formed by inserting the column of matrix B next to the columns of A , is known as **augmented matrix** of the matrices A and B . We shall denote it by $[A : B]$. The following theorem tells us about the consistency of the system $AX = B$.

Theorem: The system $AX = B$ of linear equations has a

- (i) unique solution if $\text{rank}(A) = \text{rank}([A : B]) = n$,
- (ii) infinitely many solutions if $\text{rank}(A) = \text{rank}([A : B]) < n$,
- (iii) no solution if $\text{rank}(A) \neq \text{rank}([A : B])$.

From this theorem, we deduce the following.

- If $B = O$, then obviously $\text{rank}(A) = \text{rank}([A : B])$. It implies that the homogeneous system $AX = O$ always has at least one solution. Further, it has the unique trivial solution $X = O$ if $\text{rank}(A) = n$ and infinitely many solutions if $\text{rank}(A) < n$.
- To find $\text{rank}([A : B])$, we find REF of the augmented matrix $[A : B]$. From the REF of $[A : B]$, we can immediately write the $\text{rank}(A)$. Then using the above theorem, we decide about the nature of solution of the system. In case the solution exists, it can be derived using the REF of the matrix $[A : B]$ as illustrated in the following example.

Ex. Test the consistency of the following system of equations and find the solution, if exists.

$$\begin{aligned} 2x + 3y + 4z &= 11, \\ x + 5y + 7z &= 15, \\ 3x + 11y + 13z &= 25. \end{aligned}$$

Sol. Considering the matrix form $AX = B$ of the given system, we have $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and the augmented matrix

$$[A : B] = \begin{bmatrix} 2 & 3 & 4 & 11 \\ 1 & 5 & 7 & 15 \\ 3 & 11 & 13 & 25 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_2$, we obtain

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 7 & 15 \\ 2 & 3 & 4 & 11 \\ 3 & 11 & 13 & 25 \end{bmatrix}.$$

Applying $R_2 \rightarrow R_2 + (-2)R_1$ and $R_3 \rightarrow R_3 + (-3)R_1$, we have

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & -7 & -10 & -19 \\ 0 & -4 & -8 & -20 \end{bmatrix}.$$

Applying $R_2 \rightarrow (-1/7)R_2$, we have

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 10/7 & 19/7 \\ 0 & -4 & -8 & -20 \end{bmatrix}.$$

Applying $R_3 \rightarrow R_3 + 4R_2$, we have

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 10/7 & 19/7 \\ 0 & 0 & -16/7 & -64/7 \end{bmatrix}.$$

Applying $R_3 \rightarrow (-7/16)R_3$, we have

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 10/7 & 19/7 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

This is the REF of $[A : B]$, which contains three non-zero rows. So $\text{rank}([A : B]) = 3$. Also, we see that REF of the matrix A contains three non-zero rows. So $\text{rank}(A) = 3$. Further, there are three variables in the given system. So $\text{rank}(A) = \text{rank}([A : B]) = 3$. Hence, the given system of equations is consistent and has a unique solution.

From the REF of $[A : B]$, the given system of equations is equivalent to

$$\begin{aligned} x + 5y + 7z &= 15, \\ y + (10/7)z &= 19/7, \\ z &= 4. \end{aligned}$$

From the third equation, we have $z = 4$. Inserting $z = 4$ into second equation, we obtain $y = -3$. Finally, plugging $z = 4$ and $y = -3$ into first equation, we get $x = 2$. Hence, the solution of the given system is $x = 2$, $y = -3$ and $z = 4$.

Note: In the above example, first we have found the REF of the matrix $[A : B]$. Then we have written the reduced system of equations and found the solution using back substitution. This approach is called **Gauss Elimination Method**. If we use RREF of $[A : B]$ to obtain the solution, then this approach is called **Gauss-Jordan Method**. For illustration of this method, we start with the REF of the matrix $[A : B]$ as obtained above. We have

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 10/7 & 19/7 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

Applying $R_2 \rightarrow R_2 + (-10/7)R_3$ and $R_1 \rightarrow R_1 + (-7)R_3$, we get

$$[A : B] \sim \begin{bmatrix} 1 & 5 & 0 & -13 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

Applying $R_1 \rightarrow R_1 - 5R_2$, we get

$$[A : B] \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \end{bmatrix}.$$

The RREF of $[A : B]$ yields $x = 2$, $y = -3$ and $z = 4$.

Ex. Test the consistency of the following system of equations

$$\begin{aligned} x + y + 2z + w &= 5, \\ 2x + 3y - z - 2w &= 2, \\ 4x + 5y + 3z &= 7. \end{aligned}$$

Sol. Here the augmented matrix is

$$[A : B] = \begin{bmatrix} 1 & 1 & 2 & 1 & 5 \\ 2 & 3 & -1 & -2 & 2 \\ 4 & 5 & 3 & 0 & 7 \end{bmatrix}.$$

Using suitable row transformations (you can do it), we find

$$[A : B] \sim \begin{bmatrix} 1 & 0 & 7 & 5 & 0 \\ 0 & 1 & -5 & -4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

We see that the $\text{rank}([A : B]) = 3$, but $\text{rank}(A) = 2$. So the given system of equations is inconsistent, that is, it has no solution.

Note: From the above example, you can understand that why there does not exist a solution when $\text{rank}(A) \neq \text{rank}([A : B])$. For, look at the reduced system of equations, which reads as

$$\begin{aligned} x + 7z + 5w &= 0, \\ y - 5z - 4w &= 0, \\ 0 &= 1. \end{aligned}$$

Obviously, the third equation is an absurd.

Also, notice that initially we are given three equations in four variables. Initial guess says that there would exist infinitely many solutions of the system. But we find that there does not exist even a single solution. So you should keep in mind that a system involving number of variables more than the number of equations need not to possess a solution.

Ex. Test the consistency of the following system of equations:

$$\begin{aligned} -x + 2y + z &= 1, \\ 3x - y + 2z &= 1, \\ y + \lambda z &= 1, \end{aligned}$$

where λ is a constant.

Sol. Here the augmented matrix is

$$[A : B] = \begin{bmatrix} -1 & 2 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 1 & \lambda & 1 \end{bmatrix}.$$

Using suitable row transformations (you can do it), we find

$$[A : B] \sim \begin{bmatrix} 1 & -2 & -1 & -1 \\ 0 & 1 & 1 & 4/5 \\ 0 & 0 & \lambda - 1 & 1/5 \end{bmatrix}.$$

We see that the $\text{rank}([A : B]) = 3$ irrespective of the values of λ , but $\text{rank}(A) = 2$ if $\lambda = 1$ and $\text{rank}(A) = 3$ if $\lambda \neq 1$. So the given system of equations is consistent with $\text{rank}([A : B]) = 3 = \text{rank}(A)$ when $\lambda \neq 1$, and possesses a unique solution in terms of λ . The reduced system of equations reads as

$$\begin{aligned} x - 2y - z &= -1, \\ y + z &= 4/5, \\ (\lambda - 1)z &= 1/5. \end{aligned}$$

which gives

$$z = \frac{1}{5(\lambda - 1)}, \quad y = \frac{4}{5} - \frac{1}{5(\lambda - 1)}, \quad x = \frac{3}{5} - \frac{1}{5(\lambda - 1)}.$$

Ex. Test the consistency of the following system of equations and find the solution, if exists.

$$\begin{aligned} 6x_1 - 12x_2 - 5x_3 + 16x_4 - 2x_5 &= -53, \\ -3x_1 + 6x_2 + 3x_3 - 9x_4 + x_5 &= 29, \\ -4x_1 + 8x_2 + 3x_3 - 10x_4 + x_5 &= 33. \end{aligned}$$

Sol. Here the augmented matrix is

$$[A : B] = \begin{bmatrix} 6 & -12 & -5 & 16 & -2 & -53 \\ -3 & 6 & 3 & -9 & 1 & 29 \\ -4 & 8 & 3 & -10 & 1 & 33 \end{bmatrix}.$$

Using suitable row transformations (you can do it), we find

$$[A : B] \sim \begin{bmatrix} 1 & -2 & 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -2 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix}.$$

We see that the rank($[A : B]$) = 3 = rank(A) < 5 (the number of variables in the system). So the given system of equations has infinitely many solutions. The reduced system of equations is

$$\begin{aligned} x_1 - 2x_2 + x_4 &= -4, \\ x_3 - 2x_4 &= 5, \\ x_5 &= 2. \end{aligned}$$

The second and fourth columns in RREF of $[A : B]$ do not carry the leading entries, and correspond to the variables x_2 and x_4 which we consider as independent variables. Let $x_2 = b$ and $x_4 = d$. So from the reduced system of equations, we get

$$x_1 = 2b - d - 4, \quad x_2 = b, \quad x_3 = 2d + 5, \quad x_4 = d, \quad x_5 = 2.$$

Hence, the complete solution set is

$$\{(x_1, x_2, x_3, x_4, x_5) = (2b - d - 4, b, 2d + 5, d, 2) : b, d \in R\}.$$

Row vectors, linear combinations and row space

The rows of a matrix are treated as row vectors. For example, consider the matrix

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 7 \\ 3 & 11 & 13 \end{bmatrix}.$$

Then there are three row vectors given by $[2, 3, 4]$, $[1, 5, 7]$ and $[3, 11, 13]$.

Sum of any scalar multiples of row vectors is called a linear combination while the set of all linear combinations of the row vectors is called row space of the matrix.

For example, if a , b and c are any three real numbers, then the expression

$$a[2, 3, 4] + b[1, 5, 7] + c[3, 11, 13] = [2a + b + 3c, 3a + 5b + 11c, 4a + 7b + 13c]$$

is a linear combination of the vectors $[2, 3, 4]$, $[1, 5, 7]$ and $[3, 11, 13]$ while the set

$$\{[2a + b + 3c, 3a + 5b + 11c, 4a + 7b + 13c] : a, b, c \in R\}$$

of all linear combinations of the row vectors of A is the row space of A .

Recall that two matrices are row equivalent if one can be derived from the other by applying suitable row transformation(s). The row spaces of two row equivalent matrices are always same. Afterall, row spaces are nothing but the linear combinations of the row vectors of the matrices. Consequently, the interplay of row operations gives rise to the same sets of linear combinations of row vectors, and hence the same row spaces. This fact is quite useful for determining a simple form of row space of a matrix. We know RREF of a matrix is its row equivalent matrix, and is unique as well. So we shall prefer to use RREF of the given matrix to write its row space. For example, consider the matrix

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 7 \\ 3 & 11 & 13 \end{bmatrix}.$$

Its RREF is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

So row space of A reads as

$$\{a[1, 0, 0] + b[0, 1, 0] + c[0, 0, 1] = [a, b, c] : a, b, c \in R\}.$$

Ex. Determine whether the row vector $[5, 17, -20]$ is in the row space of the matrix

$$P = \begin{bmatrix} 3 & 1 & -2 \\ 4 & 0 & 1 \\ -2 & 4 & -3 \end{bmatrix}.$$

Sol. We need to check whether there exists three real numbers a , b and c such that

$$[5, 17, -20] = a[3, 1, -2] + b[4, 0, 1] + c[-2, 4, -3].$$

This gives the following system of linear equations:

$$\begin{aligned} 3a + 4b - 2c &= 5, \\ a + 4c &= 17, \\ -2a + b - 3c &= -20. \end{aligned}$$

Here the augmented matrix is

$$[A : B] = \begin{bmatrix} 3 & 4 & -2 & 5 \\ 1 & 0 & 4 & 17 \\ -2 & 1 & -3 & -20 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

So we get $a = 5$, $b = -1$, $c = 3$, and

$$[5, 17, -20] = 5[3, 1, -2] - [4, 0, 1] + 3[-2, 4, -3].$$

Thus, $[5, 17, -20]$ is a linear combination of the row vectors of P , and hence is in the row space of P .

Linearly independent row vectors

The rows or row vectors of a matrix are said to be linearly independent (LI) if any row vector does not belong to the row space of remaining row vectors, that is, if any row vector is not the linear combination of the remaining row vectors. Equivalently, the row vectors are LI if their linear combination is zero vector only when all scalars are 0. The row vectors which are not LI, are said to be linearly dependent (LD).

To determine linear independence of the row vectors of a matrix, put the linear combination of the row vectors equal to zero vector, and determine the values of scalars. If all the scalars are 0, the vectors are LI.

Ex. Test the linear independence of the row vectors of the matrix

$$P = \begin{bmatrix} 3 & 1 & -2 \\ 4 & 0 & 1 \\ -2 & 4 & -3 \end{bmatrix}.$$

Sol. We need to find three real numbers a , b and c such that

$$a[3, 1, -2] + b[4, 0, 1] + c[-2, 4, -3] = [0, 0, 0].$$

This gives the following system of linear equations:

$$3a + 4b - 2c = 0,$$

$$a + 4c = 0,$$

$$-2a + b - 3c = 0.$$

Here the augmented matrix is

$$[A : B] = \begin{bmatrix} 3 & 4 & -2 & 0 \\ 1 & 0 & 4 & 0 \\ -2 & 1 & -3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

So we get $a = 0$, $b = 0$, $c = 0$. Thus, all the scalars are 0, which shows that the row vectors of the matrix P are LI.

Ex. Test the linear independence of the row vectors of the matrix

$$P = \begin{bmatrix} 3 & 1 & -2 \\ 4 & 0 & 1 \\ 7 & 1 & -1 \end{bmatrix}.$$

Sol. We need to find three real numbers a , b and c such that

$$a[3, 1, -2] + b[4, 0, 1] + c[7, 1, -1] = [0, 0, 0].$$

This gives the following system of linear equations:

$$3a + 4b + 7c = 0,$$

$$a + c = 0,$$

$$-2a + b - c = 0.$$

Here the augmented matrix is

$$[A : B] = \begin{bmatrix} 3 & 4 & 7 & 0 \\ 1 & 0 & 1 & 0 \\ -2 & 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

So we get $a = 0$, $b = 0$, but c is arbitrary. Thus, the row vectors of the matrix P are not LI. Notice that the third row in P is sum of the first two rows. That is why we got the linear dependence of the row vectors of P .

Note: We can talk about the linear independence of the rows of a matrix by looking at the rank of the matrix as well. If rank of a matrix is equal to the number of its rows, then the rows of the matrix are LI. At this stage, you should understand the interplay of row operations, RREF, rank, linear system of equations and linear independence of rows.

Homework: Do the problems of exercises 2.1 to 2.4 from the textbook.

Chapter 3 (3.4)

Eigenvalues and Eigenvectors

A real number λ is an eigenvalue of an n -square matrix A iff there exists a non-zero n -vector X such that $AX = \lambda X$ or $(A - \lambda I_n)X = \mathbf{0}$. The non-zero vector X is called eigenvector of A corresponding to the eigenvalue λ . Since the non-zero vector X is non-trivial solution of the homogeneous system $(A - \lambda I_n)X = \mathbf{0}$, we must have $|A - \lambda I_n| = 0$. This equation, known as the characteristic equation of A , yields eigenvalues of A . So to find the eigenvalues of A , we solve the equation $|A - \lambda I_n| = 0$.

The eigenvectors of A corresponding to λ are the **non-trivial** solutions ($X \neq \mathbf{0}$) of the homogeneous system $(A - \lambda I_n)X = \mathbf{0}$.

The set $E_\lambda = \{X : AX = \lambda X\}$ is known as the eigenspace of λ . Note that E_λ contains all eigenvectors of A corresponding to the eigenvalue λ in addition to the vector $X = \mathbf{0}$ since $A\mathbf{0} = \lambda\mathbf{0}$. Of course, by definition $X = \mathbf{0}$ is not an eigenvector of A .

Ex. Find eigenvalues and eigenvectors of $A = \begin{bmatrix} 12 & -51 \\ 2 & -11 \end{bmatrix}$.

Sol. Here, the characteristic equation of A , that is, $|A - \lambda I_2| = 0$ reads as

$$\left| \begin{bmatrix} 12 - \lambda & -51 \\ 2 & -11 - \lambda \end{bmatrix} \right| = 0.$$

This leads to a quadratic equation in λ given by

$$\lambda^2 - \lambda - 30 = 0.$$

Its roots are $\lambda = 6, -5$, the eigenvalues of A .

Now, the eigenvectors corresponding to $\lambda = 6$, are the non-trivial solutions X of the homogeneous system $(A - 6I_2)X = \mathbf{0}$. So to find eigenvectors of A corresponding to the eigenvalue $\lambda = 6$, we need to solve the homogeneous system:

$$\begin{bmatrix} 6 & -51 \\ 2 & -17 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \rightarrow (1/6)R_1$, we get

$$\begin{bmatrix} 1 & -17/2 \\ 2 & -17 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Applying $R_2 \rightarrow R_2 - 2R_1$, we get

$$\begin{bmatrix} 1 & -17/2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

So the system reduces to $x_1 - (17/2)x_2 = 0$. Letting $x_2 = a$, we get $x_1 = (17/2)a$. So

$$[x_1, x_2] = [(17/2)a, a] = (1/2)a[17, 2]$$

. So the eigenvectors corresponding to $\lambda = 6$ are non-zero multiples of the vector $[17, 2]$. The eigenspace corresponding to $\lambda = 6$, therefore, is $E_6 = \{a[17, 2] : a \in R\}$.

Likewise, to find the eigenvectors corresponding to $\lambda = -5$, we solve the homogeneous system $(A + 5I_2)X = \mathbf{0}$, that is,

$$\begin{bmatrix} 17 & -51 \\ 2 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \rightarrow (1/17)R_1$ and $R_2 \rightarrow (1/2)R_2$, we obtain

$$\begin{bmatrix} 1 & -3 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Applying $R_2 \rightarrow R_2 - R_1$, we have

$$\begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

So the system reduces to $x_1 - 3x_2 = 0$. Letting $x_2 = a$, we get $x_1 = 3a$. So

$$[x_1, x_2] = [a, 3a] = a[1, 3].$$

So the eigenvectors corresponding to $\lambda = -5$ are non-zero multiples of the vector $[3, 1]$. The eigenspace corresponding to $\lambda = -5$, therefore, is $E_{-5} = \{a[3, 1] : a \in R\}$.

Ex. Find eigenvalues and eigenvectors of $A = \begin{bmatrix} -4 & 8 & -12 \\ 6 & -6 & 12 \\ 6 & -8 & 14 \end{bmatrix}$.

Sol. Here, the characteristic equation of A , that is, $|A - \lambda I_3| = 0$ reads as

$$\begin{vmatrix} -4 - \lambda & 8 & -12 \\ 6 & -6 - \lambda & 12 \\ 6 & -8 & 14 - \lambda \end{vmatrix} = 0.$$

This leads to a cubic equation in λ given by

$$\lambda^3 - 4\lambda^2 + 4\lambda = 0.$$

Its roots are $\lambda = 0, 2, 2$, the eigenvalues of A .

Let us first find the eigenvectors of A corresponding to $\lambda = 0$. For this, we need to find non-zero solutions X of the homogeneous system $(A - 0I_3)X = 0$, that is,

$$\begin{bmatrix} -4 & 8 & -12 \\ 6 & -6 & 12 \\ 6 & -8 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Using suitable row operations, we find

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

So the reduced system of equations is

$$x_1 + x_3 = 0, \quad x_2 - x_3 = 0.$$

Letting $x_3 = a$, we get $x_1 = -a$ and $x_2 = a$. So

$$[x_1, x_2, x_3] = [-a, a, a] = a[-1, 1, 1].$$

Thus, the eigenvectors of A corresponding to $\lambda = 0$ are non-zero multiples of the vector $X_1 = [-1, 1, 1]$. The eigenspace corresponding to $\lambda = 0$, therefore, is $E_0 = \{aX_1 : a \in R\}$.

Now, let us find the eigenvectors of A corresponding to $\lambda = 2$. For this, we need to find non-zero solutions X of the homogeneous system $(A - 2I_3)X = \mathbf{0}$, that is,

$$\begin{bmatrix} -6 & 8 & -12 \\ 6 & -8 & 12 \\ 6 & -8 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Using suitable row operations, we find

$$\begin{bmatrix} 1 & -4/3 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

So the system reduces to

$$x_1 - (4/3)x_2 + 2x_3 = 0.$$

Letting $x_2 = a$ and $x_3 = b$, we get $x_1 = (4/3)a - 2b$. So

$$[x_1, x_2, x_3] = [(4/3)a - 2b, a, b] = [(4/3)a, a, 0] + [-2b, 0, b] = (1/3)a[4, 3, 0] + b[-2, 0, 1].$$

Thus, the eigenvectors corresponding to $\lambda = 2$ are non-trivial linear combinations of the vectors $X_2 = [4, 3, 0]$ and $X_3 = [-2, 0, 1]$. So $E_2 = \{aX_2 + bX_3 : a, b \in R\}$ is the eigenspace corresponding to $\lambda = 2$.

Note: The algebraic multiplicity of an eigenvalue is defined as the number of times it repeats. In the above example, the eigenvalue $\lambda = 2$ repeats two times. So its algebraic multiplicity is 2. Also, we get two linearly independent eigenvectors $X_2 = [4, 3, 0]$ and $X_3 = [-2, 0, 1]$ corresponding to $\lambda = 2$. The following example shows that there may not exist as many linearly independent eigenvectors as the algebraic multiplicity of an eigenvalue.

Ex. If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, then eigenvalues of A are $\lambda = 0, 0, 0$. The eigenvectors corresponding to $\lambda = 0$ are non-zero multiples of the vector $X = [1, 0, 0]$. The eigenspace corresponding to $\lambda = 0$, therefore, is $E_0 = \{a[1, 0, 0] : a \in R\}$. Please try this example yourself. Notice that there is only one linearly independent eigenvector $X = [1, 0, 0]$ corresponding to the repeated eigenvalue (repeating thrice) $\lambda = 0$.

Note: One can easily prove the following properties of eigen values.

- (i) Sum of eigenvalues of a matrix A is equal to the trace of A , that is, the sum of diagonal elements of A .
- (ii) Product of eigenvalues of a matrix A is equal to the determinant of A . It further implies that determinant of a matrix vanishes iff at least one eigenvalue of the matrix is 0.
- (iii) If λ is eigenvalue of A , then λ^m is eigenvalue of A^m where m is any positive integer; $1/\lambda$ is eigenvalue of inverse of A , that is, A^{-1} ; $\lambda - k$ is eigenvalue of $A - kI$, where k is any real number.

Diagonalization

A square matrix A is said to be similar to a matrix B if there exists a non-singular matrix P such that $P^{-1}AP = B$. In case, B is a diagonal matrix, we say that A is a diagonalizable matrix. Thus, a square matrix A is diagonalizable if there exists a non-singular matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix.

Suppose an n -square matrix A has n linearly independent eigenvectors $X_1, X_2, \dots, X_n$ corresponding to the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Let $P = [X_1 \ X_2 \ \dots \ X_n]$. Then we have

$$AP = [AX_1 \ AX_2 \ \dots \ AX_n] = [\lambda_1 X_1 \ \lambda_2 X_2 \ \dots \ \lambda_n X_n] = [X_1 \ X_2 \ \dots \ X_n] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} = PD.$$

This shows that if we construct P from eigenvectors of A , then A is diagonalizable, and $P^{-1}AP = D$ has eigenvalues of A at the diagonal places.

Note: If A has n different eigenvalues, then it can be proved that there exist n linearly independent eigenvectors of A and consequently A is diagonalizable. However, there may exist n linearly independent eigenvectors even if A has repeated eigenvalues as we have seen earlier. Such a matrix is also, of course, diagonalizable. In case, A does not have n linearly independent eigenvectors, it is not diagonalizable.

Ex. If $A = \begin{bmatrix} 12 & -51 \\ 2 & -11 \end{bmatrix}$, then $P = \begin{bmatrix} 17 & 3 \\ 2 & 1 \end{bmatrix}$ and $P^{-1}AP = \begin{bmatrix} 6 & 0 \\ 0 & -5 \end{bmatrix}$. (Verify!)

Ex. If $A = \begin{bmatrix} -4 & 8 & -12 \\ 6 & -6 & 12 \\ 6 & -8 & 14 \end{bmatrix}$, then $P = \begin{bmatrix} 4 & -2 & -1 \\ 3 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ and $P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. (Verify!)

Note: If A is a diagonalizable matrix, that is, $P^{-1}AP = D$ or $A = PDP^{-1}$, then for any positive integer n , we have $A^n = PD^nP^{-1}$. For,

$$A^2 = (PDP^{-1})^2 = PDP^{-1}PDP^{-1} = PD^2P^{-1}.$$

Likewise, $A^3 = PD^3P^{-1}$. So in general, $A^n = PD^nP^{-1}$.

This result can be utilized to evaluate powers of a diagonalizable matrix easily.

Ex. Determine A^2 , where $A = \begin{bmatrix} -4 & 8 & -12 \\ 6 & -6 & 12 \\ 6 & -8 & 14 \end{bmatrix}$.

So. $P = \begin{bmatrix} 4 & -2 & -1 \\ 3 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ 3 & -4 & 7 \\ -3 & 4 & -6 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

$$\text{So } A^2 = PD^2P^{-1} = \begin{bmatrix} 4 & -2 & -1 \\ 3 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 3 & -4 & 7 \\ -3 & 4 & -6 \end{bmatrix} = \begin{bmatrix} -8 & 16 & -24 \\ 12 & -12 & 24 \\ 12 & -16 & 28 \end{bmatrix}.$$

Homework: Do exercise 3.4 from the textbook.

Chapter 4 (4.1-4.7)

Before going through the concept of real vector space, you must be familiar with the following axioms (mathematical statements without proof) satisfied by the real numbers.

1. **Closure property of addition:** Sum of any two real numbers is a real number, that is, $a, b \in R$ implies $a + b \in R$. We also say that R is closed with respect to addition or real numbers satisfy closure property with respect to addition.
2. **Commutative property of addition:** Real numbers are commutative in addition, that is, $a + b = b + a$ for all $a, b \in R$.
3. **Associative property of addition:** Real numbers are associative in addition, that is, $a + (b + c) = (a + b) + c$ for all $a, b, c \in R$.
4. **Additive identity:** The real number 0 is the additive identity of real numbers, that is, $a + 0 = a = 0 + a$ for all $a \in R$.
5. **Additive inverse:** Additive inverse exists for every real number. Given any $a \in R$, we have $-a \in R$ such that $a + (-a) = 0 = (-a) + a$. So $-a$ is additive inverse of a .
6. **Closure property of multiplication:** Product of any two real numbers is a real number, that is, $a, b \in R$ implies $a.b \in R$. We also say that R is closed with respect to multiplication or real numbers satisfy closure property with respect to multiplication.
7. **Commutative property of multiplication:** Real numbers are commutative in multiplication, that is, $a.b = b.a$ for all $a, b \in R$.
8. **Associative property of multiplication:** Real numbers are associative in multiplication, that is,
 $a.(b.c) = (a.b).c$ for all $a, b, c \in R$.
9. **Multiplicative identity:** The real number 1 is the multiplicative identity of real numbers, that is, $a.1 = a = 1.a$ for all $a \in R$.
10. **Multiplicative inverse:** Multiplicative inverse exists for every non-zero real number. Given any non-zero $a \in R$, we have $1/a \in R$ such that $a.(1/a) = 1 = (1/a).a$. So $1/a$ is multiplicative inverse of a .
11. **Multiplication is distributive over addition.**
For any $a, b, c \in R$, we have $a.(b + c) = a.b + a.c$ (Left Distribution Law)
 $(b + c).a = b.a + c.a$ (Right Distribution Law).

Note: In real numbers, division is only right distributive over addition. For example,

$$(4 + 5)/7 = 4/7 + 5/7.$$

But

$$7/(4 + 5) \neq 7/4 + 7/5.$$

Real Vector Space

A non-empty set V is said to be a real vector space if there are defined two operations called vector addition and scalar multiplication denoted by $\oplus$ and $\odot$ respectively such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in R$, the following properties are satisfied:

1. $\mathbf{u} \oplus \mathbf{v} \in V$ (Closure property)
2. $\mathbf{u} \oplus \mathbf{v} = \mathbf{v} \oplus \mathbf{u}$ (Commutative property)
3. $(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = \mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w})$ (Associative property)
4. There exists some element $\mathbf{0} \in V$ such that $\mathbf{u} \oplus \mathbf{0} = \mathbf{u} = \mathbf{0} \oplus \mathbf{u}$. (Existence of additive identity)
5. There exists $-\mathbf{u} \in V$ such that $\mathbf{u} \oplus (-\mathbf{u}) = \mathbf{0} = (-\mathbf{u}) \oplus \mathbf{u}$. (Existence of additive inverse)
6. $a \odot \mathbf{u} \in V$
7. $a \odot (\mathbf{u} \oplus \mathbf{v}) = a \odot \mathbf{u} \oplus a \odot \mathbf{v}$
8. $(a + b) \odot \mathbf{u} = a \odot \mathbf{u} \oplus b \odot \mathbf{u}$
9. $(ab) \odot \mathbf{u} = a \odot (b \odot \mathbf{u})$
10. $1 \odot \mathbf{u} = \mathbf{u}$

Note: Elements of the vector space V are called as vectors while that of R are called as scalars. In what follows, a vector space shall mean a real vector space.

Note: (i) Any scalar multiplied with the zero vector gives zero vector, that is, $a \odot \mathbf{0} = \mathbf{0}$. For,

$$a \odot \mathbf{0} = a \odot \mathbf{0} \oplus \mathbf{0} = a \odot \mathbf{0} \oplus a \odot \mathbf{0} \oplus (-a \odot \mathbf{0}) = a \odot (\mathbf{0} \oplus \mathbf{0}) \oplus (-a \odot \mathbf{0}) = a \odot \mathbf{0} \oplus (-a \odot \mathbf{0}) = \mathbf{0}.$$

For clarity, let us use $+$ and $.$ symbols in place of $\oplus$ and $\odot$ respectively. Then we have

$$a.\mathbf{0} = a.\mathbf{0} + \mathbf{0} = a.\mathbf{0} + a.\mathbf{0} + (-a.\mathbf{0}) = a.(\mathbf{0} + \mathbf{0}) + (-a.\mathbf{0}) = a.\mathbf{0} + (-a.\mathbf{0}) = \mathbf{0}.$$

(ii) The scalar 0 multiplied with any vector gives zero vector, that is, $0 \odot \mathbf{u} = \mathbf{0}$. For,

$$0 \odot \mathbf{u} = (0 - 0) \odot \mathbf{u} = 0 \odot \mathbf{u} - 0 \odot \mathbf{u} = \mathbf{0}.$$

$$0.\mathbf{u} = (0 - 0).\mathbf{u} = 0.\mathbf{u} - 0.\mathbf{u} = \mathbf{0}.$$

(iii) $(-1) \odot \mathbf{u}$ gives the additive inverse $-\mathbf{u}$ of $\mathbf{u}$. For,

$$(-1) \odot \mathbf{u} \oplus \mathbf{u} = (-1) \odot \mathbf{u} \oplus 1 \odot \mathbf{u} = (-1 + 1) \odot \mathbf{u} = 0 \odot \mathbf{u} = \mathbf{0}.$$

$$(-1).\mathbf{u} + \mathbf{u} = (-1).\mathbf{u} + 1.\mathbf{u} = (-1 + 1).\mathbf{u} = 0.\mathbf{u} = \mathbf{0}.$$

Ex. The set R of all real numbers is a vector space with respect to the following operations:

- $u \oplus v = u + v$, (Vector Addition)
- $a \odot u = au$, (Scalar Multiplication)

for all $a, u, v \in R$.

Sol. In this case, $V = R$, and all the properties of vector space are easily verifiable using the axioms satisfied by the real numbers.

Note: The set $V = \{0\}$, carrying only one real number namely 0, is a real vector space with respect to the operations mentioned in the above example. Think! It is easy!

Ex. The set $R^2 = R \times R = \{[x_1, x_2] : x_1, x_2 \in R\}$ of all ordered pairs of real numbers is a vector space with respect to the following operations:

- $[x_1, x_2] \oplus [y_1, y_2] = [x_1 + y_1, x_2 + y_2]$, (Vector Addition)
- $a \odot [x_1, x_2] = [ax_1, ax_2]$, (Scalar Multiplication)

for all $a \in R$ and $[x_1, x_2], [y_1, y_2] \in R^2$.

Sol. Let $\mathbf{u} = [x_1, x_2]$, $\mathbf{v} = [y_1, y_2]$ and $\mathbf{w} = [z_1, z_2]$ be members of $V = R^2$, and a, b be any two real numbers. Then we have the following properties:

1. **Closure Property:**

$\mathbf{u} \oplus \mathbf{v} = [x_1, x_2] \oplus [y_1, y_2] = [x_1 + y_1, x_2 + y_2] \in R^2$ since $x_1 + y_1$ and $x_2 + y_2$ are real numbers.

2. **Commutative Property:**

$\mathbf{u} \oplus \mathbf{v} = [x_1, x_2] \oplus [y_1, y_2] = [x_1 + y_1, x_2 + y_2]$,

$\mathbf{v} \oplus \mathbf{u} = [y_1, y_2] \oplus [x_1, x_2] = [y_1 + x_1, y_2 + x_2]$.

But $[x_1 + y_1, x_2 + y_2] = [y_1 + x_1, y_2 + x_2]$ since real numbers are commutative in addition.

$$\therefore \mathbf{u} \oplus \mathbf{v} = \mathbf{v} \oplus \mathbf{u}.$$

3. **Associative Property:**

$(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = ([x_1, x_2] \oplus [y_1, y_2]) \oplus [z_1, z_2] = ([x_1 + y_1, x_2 + y_2]) \oplus [z_1, z_2] = [(x_1 + y_1) + z_1, (x_2 + y_2) + z_2]$

$\mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w}) = [x_1, x_2] \oplus ([y_1, y_2] \oplus [z_1, z_2]) = [x_1, x_2] \oplus ([y_1 + z_1, y_2 + z_2]) = [x_1 + (y_1 + z_1), x_2 + (y_2 + z_2)]$.

Since real numbers are associative in addition, we have

$$[(x_1 + y_1) + z_1, (x_2 + y_2) + z_2] = [x_1 + (y_1 + z_1), x_2 + (y_2 + z_2)].$$

It implies that $(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = \mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w})$.

4. **Existence of identity:**

There exists $\mathbf{0} = [0, 0] \in R^2$ such that

$\mathbf{u} \oplus \mathbf{0} = [x_1, x_2] \oplus [0, 0] = [x_1 + 0, x_2 + 0] = [x_1, x_2] = \mathbf{u}$,

$\mathbf{0} \oplus \mathbf{u} = [0, 0] \oplus [x_1, x_2] = [0 + x_1, 0 + x_2] = [x_1, x_2] = \mathbf{u}$.

So $\mathbf{u} \oplus \mathbf{0} = \mathbf{u} = \mathbf{0} \oplus \mathbf{u}$. Therefore $[0, 0]$ is additive identity in R^2 .

5. **Existence of inverse:** There exists $-\mathbf{u} = [-x_1, -x_2] \in R^2$ such that

$\mathbf{u} \oplus (-\mathbf{u}) = [x_1, x_2] \oplus [-x_1, -x_2] = [x_1 - x_1, x_2 - x_2] = [0, 0] = \mathbf{0}$,

$(-\mathbf{u}) \oplus \mathbf{u} = [-x_1, -x_2] \oplus [x_1, x_2] = [-x_1 + x_1, -x_2 + x_2] = [0, 0] = \mathbf{0}$.

So $\mathbf{u} \oplus (-\mathbf{u}) = \mathbf{0} = (-\mathbf{u}) \oplus \mathbf{u}$. This shows that $-\mathbf{u} = [-x_1, -x_2]$ is additive inverse of $\mathbf{u} = [x_1, x_2]$ in R^2 .

6. $a \odot \mathbf{u} = a \odot [x_1, x_2] = [ax_1, ax_2] \in R^2$

7. $a \odot (\mathbf{u} \oplus \mathbf{v}) = a \odot ([x_1, x_2] \oplus [y_1, y_2]) = a[x_1 + y_1, x_2 + y_2] = [a(x_1 + y_1), a(x_2 + y_2)] = [ax_1 + ay_1, ax_2 + ay_2],$
 $a \odot \mathbf{u} \oplus a \odot \mathbf{v} = a \odot [x_1, x_2] \oplus a \odot [y_1, y_2] = [ax_1, ax_2] \oplus [ay_1, ay_2] = [ax_1 + ay_1, ax_2 + ay_2].$
 $\therefore a \odot (\mathbf{u} \oplus \mathbf{v}) = a \odot \mathbf{u} \oplus a \odot \mathbf{v}.$
8. $(a + b) \odot \mathbf{u} = (a + b)[x_1, x_2] = [(a + b)x_1, (a + b)x_2] = [ax_1 + bx_1, ax_2 + bx_2],$
 $a \odot \mathbf{u} \oplus b \odot \mathbf{u} = a[x_1, x_2] \oplus b[x_1, x_2] = [ax_1, ax_2] \oplus [bx_1, bx_2] = [ax_1 + bx_1, ax_2 + bx_2].$
 $\therefore (a + b) \odot \mathbf{u} = a \odot \mathbf{u} \oplus b \odot \mathbf{u}$
9. $(ab) \odot \mathbf{u} = (ab) \odot [x_1, x_2] = [(ab)x_1, (ab)x_2],$
 $a \odot (b \odot \mathbf{u}) = a \odot (b \odot [x_1, x_2]) = a \odot [bx_1, bx_2] = [a(bx_1), a(bx_2)].$
 But $[(ab)x_1, (ab)x_2] = [a(bx_1), a(bx_2)]$ since real numbers are associative in multiplication.
 So $(ab) \odot \mathbf{u} = a \odot (b \odot \mathbf{u}).$
10. $1 \odot \mathbf{u} = 1 \odot [x_1, x_2] = [1.x_1, 1.x_2] = [x_1, x_2] = \mathbf{u}$

Hence R^2 is a real vector space.

Note: In general, the set $R^n = \{[x_1, x_2, \dots, x_n] : x_i \in R, i = 1, 2, \dots, n\}$ of all ordered n -tuples of real numbers is a vector space with respect to the following operations:

- $[x_1, x_2, \dots, x_n] \oplus [y_1, y_2, \dots, y_n] = [x_1 + y_1, x_2 + y_2, \dots, x_n + y_n],$ (Vector Addition)
- $a \odot [x_1, x_2, \dots, x_n] = [ax_1, ax_2, \dots, ax_n],$ (Scalar Multiplication)

for all $a \in R$ and $[x_1, x_2, \dots, x_n], [y_1, y_2, \dots, y_n] \in R^n.$

Ex. The set $M_{mn} = \{[a_{ij}]_{m \times n} : a_{ij} \in R\}$ of all $m \times n$ matrices with real entries is a vector space with respect to the following operations:

- $[a_{ij}]_{m \times n} \oplus [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n},$ (Vector Addition)
- $a \odot [a_{ij}]_{m \times n} = [aa_{ij}]_{m \times n},$ (Scalar Multiplication)

for all $a \in R$ and $[a_{ij}]_{m \times n}, [b_{ij}]_{m \times n} \in M_{mn}.$ Notice that vector addition is usual addition of matrices, and scalar multiplication is the usual scalar multiplication in matrices.

Sol. Let $\mathbf{u} = [a_{ij}]_{m \times n}, \mathbf{v} = [b_{ij}]_{m \times n}$ and $\mathbf{w} = [c_{ij}]_{m \times n}$ be members of $V = M_{mn},$ and a, b be any two real numbers. Then we have the following properties:

1. Closure Property:

$$\mathbf{u} \oplus \mathbf{v} = [a_{ij}]_{m \times n} \oplus [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n} \in M_{mn}.$$

2. Commutative Property:

$$\mathbf{u} \oplus \mathbf{v} = [a_{ij}]_{m \times n} \oplus [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n}.$$

$$\mathbf{v} \oplus \mathbf{u} = [b_{ij}]_{m \times n} \oplus [a_{ij}]_{m \times n} = [b_{ij} + a_{ij}]_{m \times n}.$$

Since $a_{ij} + b_{ij} = b_{ij} + a_{ij},$ so we have $\mathbf{u} \oplus \mathbf{v} = \mathbf{v} \oplus \mathbf{u}.$

3. Associative Property:

$$(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = ([a_{ij}]_{m \times n} \oplus [b_{ij}]_{m \times n}) \oplus [c_{ij}]_{m \times n} = ([a_{ij} + b_{ij}]_{m \times n}) \oplus [c_{ij}]_{m \times n} = [(a_{ij} + b_{ij}) + c_{ij}]_{m \times n},$$

$$\mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w}) = [a_{ij}]_{m \times n} \oplus ([b_{ij}]_{m \times n} \oplus [c_{ij}]_{m \times n}) = [a_{ij}]_{m \times n} \oplus [b_{ij} + c_{ij}]_{m \times n} = [a_{ij} + (b_{ij} + c_{ij})]_{m \times n}.$$

Since $(a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij}),$ so we get $(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = \mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w}).$

4. Existence of identity:

There exists $\mathbf{0} = [0]_{m \times n} \in M^{mn}$ such that

$$\mathbf{u} \oplus \mathbf{0} = [a_{ij}]_{m \times n} \oplus [0]_{m \times n} = [a_{ij} + 0]_{m \times n} = [a_{ij}]_{m \times n} = \mathbf{u},$$

$$\mathbf{0} \oplus \mathbf{u} = [0]_{m \times n} \oplus [a_{ij}]_{m \times n} = [0 + a_{ij}]_{m \times n} = [a_{ij}]_{m \times n} = \mathbf{u}.$$

So $\mathbf{u} \oplus \mathbf{0} = \mathbf{u} = \mathbf{0} \oplus \mathbf{u}.$ Therefore, $[0]_{m \times n},$ the null matrix of order $m \times n$ is the additive identity in $M_{mn}.$

5. **Existence of inverse:** There exists $-\mathbf{u} = [-a_{ij}]_{m \times n} \in M_{mn}$ such that
 $\mathbf{u} \oplus (-\mathbf{u}) = [a_{ij}]_{m \times n} \oplus [-a_{ij}]_{m \times n} = [a_{ij} - a_{ij}]_{m \times n} = [0]_{m \times n} = \mathbf{0}$,
 $(-\mathbf{u}) \oplus \mathbf{u} = [-a_{ij}]_{m \times n} \oplus [a_{ij}]_{m \times n} = [-a_{ij} + a_{ij}]_{m \times n} = [0]_{m \times n} = \mathbf{0}$.
 So $\mathbf{u} \oplus (-\mathbf{u}) = \mathbf{0} = (-\mathbf{u}) \oplus \mathbf{u}$. This shows that $-\mathbf{u} = [-a_{ij}]_{m \times n}$ is additive inverse of $\mathbf{u} = [a_{ij}]_{m \times n}$ in M_{mn} .
6. $a \odot \mathbf{u} = a \odot [a_{ij}]_{m \times n} = [aa_{ij}]_{m \times n} \in M_{mn}$
7. $a \odot (\mathbf{u} \oplus \mathbf{v}) = a \odot ([a_{ij}]_{m \times n} \oplus [b_{ij}]_{m \times n}) = a \odot [a_{ij} + b_{ij}]_{m \times n} = [a(a_{ij} + b_{ij})]_{m \times n} = [aa_{ij} + ab_{ij}]_{m \times n}$,
 $a \odot \mathbf{u} \oplus a \odot \mathbf{v} = a \odot [a_{ij}]_{m \times n} \oplus a \odot [b_{ij}]_{m \times n} = [aa_{ij}]_{m \times n} \oplus [ab_{ij}]_{m \times n} = [aa_{ij} + ab_{ij}]_{m \times n}$.
 $\therefore a \odot (\mathbf{u} \oplus \mathbf{v}) = a \odot \mathbf{u} \oplus a \odot \mathbf{v}$.
8. $(a + b) \odot \mathbf{u} = (a + b) \odot [a_{ij}]_{m \times n} = [(a + b)a_{ij}]_{m \times n} = [aa_{ij} + ba_{ij}]_{m \times n}$,
 $a \odot \mathbf{u} \oplus b \odot \mathbf{u} = a \odot [a_{ij}]_{m \times n} \oplus b \odot [a_{ij}]_{m \times n} = [aa_{ij}]_{m \times n} \oplus [ba_{ij}]_{m \times n} = [aa_{ij} + ba_{ij}]_{m \times n}$.
 $\therefore (a + b) \odot \mathbf{u} = a \odot \mathbf{u} \oplus b \odot \mathbf{u}$.
9. $(ab) \odot \mathbf{u} = (ab) \odot [a_{ij}]_{m \times n} = [(ab)a_{ij}]_{m \times n}$,
 $a \odot (b \odot \mathbf{u}) = a \odot (b \odot [a_{ij}]_{m \times n}) = a \odot [ba_{ij}]_{m \times n} = [a(ba_{ij})]_{m \times n}$.
 But $[(ab)a_{ij}]_{m \times n} = [a(ba_{ij})]_{m \times n}$ since real numbers are associative in multiplication.
 So $(ab) \odot \mathbf{u} = a \odot (b \odot \mathbf{u})$.
10. $1 \odot \mathbf{u} = 1 \odot [a_{ij}]_{m \times n} = [1.a_{ij}]_{m \times n} = [a_{ij}]_{m \times n} = \mathbf{u}$

Hence M_{mn} is a real vector space.

Ex. The set $P_n = \left\{ \sum_{i=0}^n a_i x^i = a_0 + a_1 x + \dots + a_n x^n : a_i \in R \right\}$ of all polynomials in x of degree $\leq n$ with real coefficients is a vector space with respect to the following operations:

- $\sum_{i=0}^n a_i x^i \oplus \sum_{i=0}^n b_i x^i = \sum_{i=0}^n (a_i + b_i) x^i$, (Vector Addition)
- $a \odot \sum_{i=0}^n a_i x^i = \sum_{i=0}^n aa_i x^i$, (Scalar Multiplication)

for all $a \in R$ and $\sum_{i=0}^n a_i x^i \in P_n$. Notice that vector addition is usual addition of polynomials, and scalar multiplication is the usual scalar multiplication in polynomials.

Sol. Please do yourself following the procedure given in the previous example(s).

Ex. The set $\Phi = \{f : f \text{ is well defined real valued function on } [0, 1]\}$ is a vector space with respect to the following operations:

- $f \oplus g = f + g$, (Vector Addition)
- $a \odot f = af$, (Scalar Multiplication)

for all $a \in R$ and $f, g \in \Phi$.

Sol. Please do yourself following the procedure given in the previous example(s).

Ex. Show that the set $\Phi = \{f : f \text{ is well defined real valued function on } [0, 1] \text{ and } f(1/2) = 1\}$ is not a vector space with respect to the following operations:

- $f \oplus g = f + g$, (Vector Addition)
- $a \odot f = af$, (Scalar Multiplication)

for all $a \in R$ and $f, g \in \Phi$.

Sol. Let $f, g \in \Phi$. Then by definition of Φ , we have
 $f(1/2) = 1, g(1/2) = 1$.

Next by definition of vector addition, we find

$$(f \oplus g)(1/2) = (f + g)(1/2) = f(1/2) + g(1/2) = 1 + 1 = 2 \neq 1.$$

So $f \oplus g \notin \Phi$. It means vector addition fails in Φ . Consequently, it is not a vector space.

Ex. The set R^+ of positive real numbers is a vector space with respect to the operations:

- $u \oplus v = uv$ (vector addition)
- $a \odot u = u^a$ (scalar multiplication)

for all $a \in R$ and $u, v \in R^+$. Here vector addition of u and v is defined by their product while scalar multiplication of a and u is defined by u raised to the power a . So be careful while verifying the properties.

Sol. Let u, v and w be members of $V = R^+$, and a, b be any two real numbers. Note that u, v and w are positive real numbers while a and b are any real numbers. Then we have the following properties:

1. **Closure Property:**

$$u \oplus v = uv \in R^+.$$

2. **Commutative Property:**

$$u \oplus v = uv = vu = v \oplus u.$$

3. **Associative Property:**

$$(u \oplus v) \oplus w = (uv) \oplus w = (uv)w = u(vw) = u \oplus (vw) = u \oplus (v \oplus w).$$

4. **Existence of identity:**

$1 \in R^+$ (denoting the positive real number 1 by $\mathbf{0}$) such that

$$u \oplus \mathbf{0} = u \oplus 1 = u.1 = u,$$

$$\mathbf{0} \oplus u = 1 \oplus u = 1.u = u.$$

So $u \oplus \mathbf{0} = u = \mathbf{0} \oplus u$. Therefore, the positive real number 1 is the additive identity in R^+ .

5. **Existence of inverse:**

Since u is a positive real number so $1/u$ is also a positive real number (denoting $1/u$ by $-u$) such that

$$u \oplus (-u) = u \oplus (1/u) = u.(1/u) = 1 = \mathbf{0},$$

$$(-u) \oplus u = (1/u) \oplus u = (1/u).u = 1 = \mathbf{0}.$$

So $u \oplus (-u) = \mathbf{0} = (-u) \oplus u$. This shows that $-u = 1/u$ is additive inverse of u in R^+ .

6. $a \odot u = u^a \in R^+$

$$7. a \odot (u \oplus v) = a \odot (uv) = (uv)^a = u^a v^a = u^a \oplus v^a = a \odot u \oplus a \odot v.$$

$$8. (a + b) \odot u = u^{(a+b)} = u^a u^b = u^a \oplus u^b = a \odot u \oplus b \odot u.$$

$$9. (ab) \odot u = u^{(ab)} = (u^b)^a = a \odot (u^b) = a \odot (b \odot u).$$

$$10. 1 \odot u = u^1 = u.$$

Hence R^+ is a real vector space.

Ex. Show that the set R of real numbers is a vector space with respect to the operations:

- $u \oplus v = (u^5 + v^5)^{1/5}$
- $a \odot u = a^{1/5}u$

for all $a, u, v \in R$. Further, the principal fifth root is to be considered in both the operations.

Sol. Let u, v and w be members of $V = R$, and a, b be any two real numbers. Then we have the following properties:

1. **Closure Property:**

$$u \oplus v = (u^5 + v^5)^{1/5} \in R.$$

2. **Commutative Property:**

$$u \oplus v = (u^5 + v^5)^{1/5} = (v^5 + u^5)^{1/5} = v \oplus u.$$

3. **Associative Property:**

$$(u \oplus v) \oplus w = ((u^5 + v^5)^{1/5}) \oplus w = ((u^5 + v^5) + w^5)^{1/5} = (u^5 + (v^5 + w^5))^{1/5} = u \oplus (v \oplus w) = u \oplus (v \oplus w).$$

4. **Existence of identity:**

$0 \in R$ (denoting the real number 0 by $\mathbf{0}$) such that

$$u \oplus \mathbf{0} = u \oplus 0 = (u^5 + 0^5)^{1/5} = u,$$

$$\mathbf{0} \oplus u = 0 \oplus u = (0^5 + u^5)^{1/5} = u.$$

So $u \oplus \mathbf{0} = u = \mathbf{0} \oplus u$. Therefore, the real number 0 is the additive identity in R .

5. **Existence of inverse:**

$-u \in R$ such that

$$u \oplus (-u) = (u^5 - u^5)^{1/5} = 0 = \mathbf{0},$$

$$(-u) \oplus u = (-u^5 + u^5)^{1/5} = 0 = \mathbf{0}.$$

So $u \oplus (-u) = \mathbf{0} = (-u) \oplus u$. This shows that $-u$ is additive inverse of u in R .

$$6. a \odot u = a^{1/5}u \in R$$

$$7. a \odot (u \oplus v) = a \odot ((u^5 + v^5)^{1/5}) = a^{1/5}(u^5 + v^5)^{1/5} = (au^5 + av^5)^{1/5} = (a^{1/5}u)^5 \oplus (a^{1/5}v)^5 = a \odot u \oplus a \odot v.$$

$$8. (a + b) \odot u = (a + b)^{1/5}u = (au^5 + bu^5)^{1/5} = (a^{1/5}u)^5 \oplus (b^{1/5}u)^5 = a \odot u \oplus b \odot u.$$

$$9. (ab) \odot u = (ab)^{1/5}u = (a^{1/5}b^{1/5})u = a^{1/5}(b^{1/5}u) = a^{1/5}(b \odot u) = a \odot (b \odot u).$$

$$10. 1 \odot u = 1^{1/5}u = u.$$

Hence R is a real vector space.

Ex. Show that the set R of real numbers is not a vector space with respect to the operations:

- $u \oplus v = (u^5 + v^5)^{1/5}$ (vector addition)
- $a \odot u = au$ (scalar multiplication)

for all $a, u, v \in R$. Further, the principal fifth root is to be considered in vector addition.

Sol. Property 8 is not satisfied. Please do yourself.

Homework: Do exercise 4.1 from the textbook.

Subspace

If V is a vector space and W is a subset of V such that W is also a vector space under the same operations as in V , then W is called a subspace of V .

Ex. The set $W = \{[x_1, 0] : x_1 \in R\}$ is a vector space under the operations of vector addition and scalar multiplication as we considered earlier in R^2 . Also, W is subset of R^2 . So W is subspace of R^2 .

Note: For checking W to be a subspace of a vector space V , we need to verify 10 properties of vector space in W . The following theorem suggests that verification of two properties is enough.

Theorem: A subset W of a vector space V is a subspace of V if and only if W is closed with respect to vector addition and scalar multiplication.

Proof: First consider that W is subspace of V . Then by definition of subspace, W is a vector space. So by definition of vector space, W is closed with respect to vector addition and scalar multiplication.

Conversely assume that the subset W of the vector space V is closed with respect to vector addition and scalar multiplication. So $\mathbf{u} \oplus \mathbf{v} \in W$ and $a \odot \mathbf{u} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$ and $a \in R$. Choosing $a = 0$, we get $0 \odot \mathbf{u} \in W$. But $0 \odot \mathbf{u} = \mathbf{0}$. So $\mathbf{0} \in W$, that is, additive identity exists in W . Again, choosing $a = -1$, we get $(-1) \odot \mathbf{u} = -\mathbf{u} \in W$. So additive inverse exists in W . Commutative and associative properties with regard to vector addition, and all the properties related to scalar multiplication shall follow in W for two reasons (i) members of W are members from V , and (ii) V is a vector space.

Note: It is also easy to show that a subset W of a vector space V is subspace of V if and only if $a \odot \mathbf{u} \oplus b \odot \mathbf{v} \in W$ for all $a, b \in R$ and $\mathbf{u}, \mathbf{v} \in W$.

Ex. Show that the set $W = \{[x_1, 0] : x_1 \in R\}$ is subspace of R^2 .

Sol. Let $\mathbf{u} = [x_1, 0]$, $\mathbf{v} = [y_1, 0] \in W$ and $a \in R$. Then we have

$$\mathbf{u} \oplus \mathbf{v} = [x_1, 0] \oplus [y_1, 0] = [x_1 + y_1, 0] \in W,$$

$$a \odot \mathbf{u} = a \odot [x_1, 0] = [ax_1, 0] \in W.$$

This shows that W is closed with respect to vector addition and scalar multiplication. Hence W is a subspace of R^2 .

Ex. The set $W = \{(a, b, a + 2b) : a, b \in R\}$ is a subspace of R^3 .

Sol. Please do yourself.

Ex. Verify whether $W = \{[x, y] : x - y = 0, x, y \in R\}$ is a subspace of R^2 .

Sol. Let $\mathbf{u} = [x_1, x_2]$, $\mathbf{v} = [y_1, y_2] \in W$ and $a \in R$. Then $x_1 - x_2 = 0$ and $y_1 - y_2 = 0$. we have

$$\mathbf{u} \oplus \mathbf{v} = [x_1, x_2] \oplus [y_1, y_2] = [x_1 + y_1, x_2 + y_2] \in W,$$

since $(x_1 + y_1) - (x_2 + y_2) = x_1 - x_2 + y_1 - y_2 = 0$.

$$a \odot \mathbf{u} = a \odot [x_1, x_2] = [ax_1, ax_2] \in W,$$

since $ax_1 - ax_2 = a(x_1 - x_2) = a \cdot 0 = 0$. This shows that W is closed with respect to vector addition and scalar multiplication. Hence W is a subspace of R^2 .

Ex. Verify whether $W = \{[x, y] : y = x^2, x, y \in R\}$ is a subspace of R^2 .

Sol. We have $[1, 1], [2, 4] \in W$ but

$$[1, 1] \oplus [2, 4] = [1 + 2, 1 + 4] = [3, 5] \notin W,$$

since $5 \neq 3^2$. So W is not a subspace of R^2 .

Ex. Verify whether $W = \left\{ \begin{bmatrix} p & q \\ r & s \end{bmatrix} : ps - qr \neq 0, p, q, r, s \in R \right\}$ is a subspace of M_{22} .

Sol. We have $\mathbf{u} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \in W$ but

$$\mathbf{u} \oplus \mathbf{v} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \oplus \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1-1 & 0+0 \\ 0+0 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin W,$$

since $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is a singular matrix. So W is not a subspace of M_{22} .

Ex. If A is n -square matrix and λ is an eigen value of A , then the eigen space $E_\lambda = \{X : AX = \lambda X\}$ of λ is a subspace of R^n .

Sol. Let $a \in R$ and $X_1, X_2 \in E_\lambda$. Then

$$A(X_1 + X_2) = AX_1 + AX_2 = \lambda X_1 + \lambda X_2 = \lambda(X_1 + X_2). \text{ So } X_1 + X_2 \in E_\lambda.$$

$$\text{Also, } A(aX_1) = aAX_1 = a\lambda X_1 = \lambda(aX_1). \text{ So } aX_1 \in E_\lambda.$$

Thus, E_λ is a subspace of R^n .

Note: If V is any vector space, then $\{0\}$ and V are its trivial subspaces. Any other subspace is called a proper subspace of V .

Ex. Let W_1 and W_2 be two subspaces of a vector space V .

- (i) Show that $W_1 \cap W_2$ is a subspace of V .
- (ii) Give an example to show that $W_1 \cup W_2$ need not be a subspace of V .
- (iii) Show that $W_1 \cup W_2$ is subspace of V iff either $W_1 \subset W_2$ or $W_2 \subset W_1$.

Sol. (i) Since W_1 and W_2 are subspaces of V . So at least the zero vector $\mathbf{0}$ lies in $W_1 \cap W_2$. So $W_1 \cap W_2 \neq \phi$. Let $\mathbf{u}, \mathbf{v} \in W_1 \cap W_2$, and $a, b \in R$. Then $a \odot \mathbf{u} \oplus b \odot \mathbf{v} \in W_1$ and $a \odot \mathbf{u} \oplus b \odot \mathbf{v} \in W_2$ since W_1 and W_2 both are subspaces of V . It follows that $a \odot \mathbf{u} \oplus b \odot \mathbf{v} \in W_1 \cap W_2$. This shows that $W_1 \cap W_2$ is a subspace of V .

(ii) $W_1 = \{[x_1, 0] : x_1 \in R\}$ and $W_2 = \{[0, x_2] : x_2 \in R\}$ both are subspaces of R^2 . Also, $[1, 0], [0, 1] \in W_1 \cup W_2$. But $[1, 0] \oplus [0, 1] = [1 + 0, 0 + 1] = [1, 1] \notin W_1 \cup W_2$. So $W_1 \cup W_2$ is not closed with respect to vector addition, and consequently it is not a subspace of R^2 .

(iii) If $W_1 \subset W_2$ or $W_2 \subset W_1$, then $W_1 \cup W_2 = W_2$ or $W_1 \cup W_2 = W_1$. But W_1 and W_2 both are subspaces of V . Thus, in both cases $W_1 \cup W_2$ is a subspace of V .

Conversely assume that $W_1 \cup W_2$ is a subspace of V . We need to prove that either $W_1 \subset W_2$ or $W_2 \subset W_1$. On the contrary assume that neither $W_1 \subset W_2$ nor $W_2 \subset W_1$. So there must exist some $\mathbf{u} \in W_1$ and $\mathbf{v} \in W_2$ such that $\mathbf{u} \notin W_2$ and $\mathbf{v} \notin W_1$.

Now $\mathbf{u}, \mathbf{v} \in W_1 \cup W_2$. So $\mathbf{u} \oplus \mathbf{v} \in W_1 \cup W_2$, by closure property of vector addition in $W_1 \cup W_2$. It implies that either $\mathbf{u} \oplus \mathbf{v} \in W_1$ or $\mathbf{u} \oplus \mathbf{v} \in W_2$. Let $\mathbf{u} \oplus \mathbf{v} \in W_1$. Also, $-\mathbf{u} \in W_1$. So by closure property of vector addition in W_1 , we have $-\mathbf{u} \oplus (\mathbf{u} \oplus \mathbf{v}) = \mathbf{v} \in W_1$, a contradiction since $\mathbf{v} \notin W_1$. Thus, either $W_1 \subset W_2$ or $W_2 \subset W_1$.

Note: Hereafter, we shall use the symbols $+$ and $\cdot$ for vector addition $\oplus$ and scalar multiplication $\odot$ respectively in all vector spaces.

Homework: Do exercise 4.2 from the textbook.

Span of a Set

Let S be any subset of a vector space V . Then the set of all linear combinations of finite number of members of S is called the span of S , and is denoted by $\text{span}(S)$ or $L(S)$. Therefore, we have

$$L(S) = \text{span}(S) = \{a_1v_1 + a_2v_2 + \dots + a_nv_n : a_i \in R, v_i \in S, i = 1, 2, \dots, n\}.$$

Ex. If $S = \{[1, 0], [0, 1]\}$, then $L(S) = \{a(1, 0) + b(0, 1) : a, b \in R\} = \{(a, b) : a, b \in R\} = R^2$.

Note: Recall that row space of a matrix is the set of all sums of scalar multiples of the row vectors of the matrix. So row space of a matrix is nothing but the span of its row vectors. Also, we know that the row spaces of two row equivalent matrices are same. This fact can be used to simplify the span of subsets of the vector spaces R^n , P_n and M_{mn} as illustrated in the following examples.

Ex. Show that span of the set $S = \{[2, 3, 4], [1, 5, 7], [3, 11, 13]\}$ is R^3 .

Sol. By definition, span of the given set S reads as

$$L(S) = \{a[2, 3, 4] + b[1, 5, 7] + c[3, 11, 13] : a, b, c \in R\} = \{[2a+b+3c, 3a+5b+11c, 4a+7b+13c] : a, b, c \in R\}.$$

For simplified span, we use suitable row operations, and find

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 7 \\ 3 & 11 & 13 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$\therefore L(S) = \{a[1, 0, 0] + b[0, 1, 0] + c[0, 0, 1] : a, b, c \in R\} = \{[a, b, c] : a, b, c \in R\} = R^3.$$

Ex. Find simplified span of the set $S = \{x^3 - 1, x^2 - x, x - 1\}$ in P_{3x} .

Sol. By definition, span of the given set S reads as

$$L(S) = \{a(x^3 - 1) + b(x^2 - x) + c(x - 1) : a, b, c \in R\} = \{ax^3 + bx^2 + cx - a - c : a, b, c \in R\}.$$

For simplified span, we write coefficients of the polynomials as row vectors, and then use suitable row operations. We find

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

$$\therefore L(S) = \{a(x^3 - 1) + b(x^2 - 1) + c(x - 1) : a, b, c \in R\} = \{ax^3 + bx^2 + cx - a - b - c : a, b, c \in R\}.$$

Ex. Find simplified span of the set $S = \left\{ \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ in M_{22} .

Sol. By definition, span of the given set S reads as

$$L(S) = \left\{ a \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} + c \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} : a, b, c \in R \right\} = \left\{ \begin{bmatrix} -a-c & a \\ b & -b+c \end{bmatrix} : a, b, c \in R \right\}$$

For simplified span, we write each matrix as continuous row vector (writing both rows of each matrix in one row), and then use suitable row operations. We find

$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

$$\therefore L(S) = \left\{ a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} : a, b, c \in R \right\} = \left\{ \begin{bmatrix} a & b \\ c & -a-b-c \end{bmatrix} : a, b, c \in R \right\}.$$

Ex. Span of the empty set is defined to be the singleton set containing the zero vector., that is, $\text{span}(\phi) = \{0\}$.

Theorem: Let S be subset of a vector space V . Then prove the following:

- (i) $L(S)$ is a subset of V
- (ii) $L(S)$ is a subspace of V
- (iii) $L(S)$ is the minimal subspace of V containing S .

Proof: (i) Let $u \in L(S)$. Then

$$u = c_1 u_1 + c_2 u_2 + \dots + c_m u_m$$

for some $c_i \in R$ and $u_i \in S$ where $i = 1, 2, \dots, m$. Now, $c_1 \in R$ and $u_1 \in S \subset V$. So $c_1 u_1 \in V$, by property 6 (scalar multiplication) of vector space. Likewise $c_2 u_2 \in V$. It follows that $c_1 u_1 + c_2 u_2 \in V$, by property 1 (vector addition) of vector space. Thus, repeated use of scalar multiplication and vector addition properties of vector space yields

$$u = c_1 u_1 + c_2 u_2 + \dots + c_m u_m \in V.$$

Therefore, $L(S)$ is subset of V .

(ii) Let $u, v \in L(S)$ and $a \in R$. Then u and v both are linear combinations of finite number of members of S .

$$\therefore u = c_1 u_1 + c_2 u_2 + \dots + c_m u_m, \quad v = d_1 v_1 + d_2 v_2 + \dots + d_n v_n,$$

for some $c_i, d_j \in R$ and $u_i, v_j \in S$ where $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$. It follows that

$$u + v = c_1 u_1 + c_2 u_2 + \dots + c_m u_m + d_1 v_1 + d_2 v_2 + \dots + d_n v_n \in L(S),$$

$$au = (ac_1)u_1 + (ac_2)u_2 + \dots + (ac_m)u_m \in L(S).$$

This shows that $L(S)$ is closed with respect to vector addition and scalar multiplication. Also, by part (i), $L(S)$ is subset of V . So $L(S)$ is subspace of V .

(iii) For any $u \in S$, we can write $u = 1.u$. So every member of S can be written as a linear combination of members of S . So $S \subset L(S)$. In part (ii), we have shown that $L(S)$ is a subspace of V . To show that $L(S)$ is minimal subspace of V containing S , it requires to prove that $L(S) \subset W$ where W is any subspace of V containing S .

Let $u \in L(S)$. Then

$$u = c_1u_1 + c_2u_2 + \dots + c_mu_m$$

for some $c_i \in R$ and $u_i \in S$ where $i = 1, 2, \dots, m$. Now W contains S . Also W being a subspace of V is a vector space. So by the properties of scalar multiplication and vector addition, it follows that

$$u = c_1u_1 + c_2u_2 + \dots + c_mu_m \in W.$$

Thus, $L(S) \subset W$. This completes the proof.

Homework: Do exercise 4.3 from the textbook.

Linear Independence

A finite non-empty subset $S = \{v_1, v_2, \dots, v_n\}$ of a vector space V is said to be linearly dependent (LD) if and only if there exist real numbers $a_1, a_2, \dots, a_n$ not all zero such that $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$.

If S is not LD, that is, $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ implies $a_1 = 0, a_2 = 0, \dots, a_n = 0$, then S is said to be linearly independent (LI).

Ex. The set $S = \{[1, 0], [0, 1]\}$ is LI in R^2 .

Sol. Here we have two vectors $v_1 = [1, 0]$ and $v_2 = [0, 1]$. To check LI/LD, we let

$$a_1v_1 + a_2v_2 = 0.$$

$$\implies a_1[1, 0] + a_2[0, 1] = [0, 0].$$

$$\implies [a_1, 0] + [0, a_2] = [0, 0].$$

$$\implies [a_1, a_2] = [0, 0].$$

$$\implies a_1 = 0, a_2 = 0.$$

Thus, the set $S = \{[1, 0], [0, 1]\}$ is LI in R^2 .

Ex. The set $S = \{[1, 2], [2, 4]\}$ is LD in R^2 .

Sol. Here we have two vectors $v_1 = [1, 2]$ and $v_2 = [2, 4]$. To check LI/LD, we let

$$a_1v_1 + a_2v_2 = 0.$$

$$\implies a_1[1, 2] + a_2[2, 4] = [0, 0].$$

$$\implies [a_1, 2a_1] + [2a_2, 4a_2] = [0, 0].$$

$$\implies [a_1 + 2a_2, 2a_1 + 4a_2] = [0, 0].$$

$$\implies a_1 + 2a_2 = 0, 2a_1 + 4a_2 = 0.$$

$$\implies a_1 + 2a_2 = 0.$$

We see a non-trivial solution, $a_1 = 2$, $a_2 = -1$. Thus, the set $S = \{[1, 2], [2, 4]\}$ is LD in R^2 .

Note: In fact, $a_1 + 2a_2 = 0$ has infinitely many non-trivial solutions. But for LD, existence of one non-trivial is sufficient.

Theorem: Two vectors in a vector space are LD if and only if one vector is scalar multiple of the other.

Proof: Let v_1 and v_2 be two vectors of a vector space V . Suppose v_1 and v_2 are LD. Then there exists real numbers a_1 and a_2 such that at least one of a_1 and a_2 is non-zero (say $a_1 \neq 0$), and

$$a_1 v_1 + a_2 v_2 = 0.$$

Since $a_1 \neq 0$, we have

$$v_1 = \left(-\frac{a_2}{a_1}\right) v_2 = 0.$$

This shows that v_1 is scalar multiple of v_2 .

Conversely assume that v_1 is scalar multiple of v_2 , that is, $v_1 = \alpha v_2$ for some real number α . Then we have,

$$(-1)v_1 + \alpha v_2 = 0.$$

We see that the linear combination of v_1 and v_2 is 0, where the scalar -1 with v_1 is non-zero. So v_1 and v_2 are LD.

Ex. Verify whether the set $S = \{[3, 1, -1], [-5, -2, 2], [2, 2, -1]\}$ is LI in R^3 .

Sol. Here we have three vectors $v_1 = [3, 1, -1]$, $v_2 = [-5, -2, 2]$ and $v_3 = [2, 2, -1]$. To check LI/LD, we let

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = 0.$$

$$\implies a_1 [3, 1, -1] + a_2 [-5, -2, 2] + a_3 [2, 2, -1] = [0, 0, 0].$$

$$\implies [3a_1 - 5a_2 + 2a_3, a_1 - 2a_2 + 2a_3, -a_1 + 2a_2 - a_3] = [0, 0, 0].$$

This gives the following homogeneous system of equations:

$$3a_1 - 5a_2 + 2a_3 = 0,$$

$$a_1 - 2a_2 + 2a_3 = 0,$$

$$-a_1 + 2a_2 - a_3 = 0.$$

Here the augmented matrix is

$$[A : B] = \begin{bmatrix} 3 & -5 & 2 & 0 \\ 1 & -2 & 2 & 0 \\ -1 & 2 & -1 & 0 \end{bmatrix}.$$

Using suitable row transformations, we find

$$[A : B] \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

So we get the trivial solution, $a_1 = 0$, $a_2 = 0$ and $a_3 = 0$.

Thus, the set $S = \{[3, 1, -1], [-5, -2, 2], [2, 2, -1]\}$ is LI in R^3 .

Note: Row reduction approach can also be applied to test the linear independence of the vectors of R^n . Just write the given vectors as the rows of a matrix say A , and find rank of A . If rank of A is equal to the number of rows or vectors, then the given vectors are LI. For example, consider the set $S = \{[3, 1, -1], [-5, -2, 2], [2, 2, -1]\}$. The matrix with rows as the vectors of S reads as

$$A = \begin{bmatrix} 3 & 1 & -1 \\ -5 & -2 & 2 \\ 2 & 2 & -1 \end{bmatrix}.$$

Using suitable row operations, we find

$$A = \begin{bmatrix} 3 & 1 & -1 \\ -5 & -2 & 2 \\ 2 & 2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

So rank of A is 3. It implies that the set $S = \{[3, 1, -1], [-5, -2, 2], [2, 2, -1]\}$ is LI.

Ex. Any set containing zero vector is always LD while a singleton set containing a non-zero vector is LI.

Sol. Let $S = \{\mathbf{0}, v_1, v_2, \dots, v_n\}$ be a set containing $\mathbf{0}$ vector. Then the expression

$$1 \cdot \mathbf{0} + 0v_1 + 0v_2 + \dots + 0v_n = \mathbf{0}$$

shows that S is LD.

Next, consider a set $A = \{v_1\}$ carrying a single non-zero vector. Then $a_1 v_1 = \mathbf{0}$ gives $a_1 = 0$ since $v_1 \neq \mathbf{0}$. So A is LI.

Theorem: A finite set S containing at least two vectors is LD iff some vector in S can be expressed as a linear combination of the other vectors in S .

Proof: Let $S = \{v_1, v_2, \dots, v_n\}$ be a set containing at least two vectors. Suppose S is LD. Then there exist real numbers $a_1, a_2, \dots, a_n$ not all zero (say some $a_m \neq 0$) such that

$$a_1 v_1 + a_2 v_2 + \dots + a_{m-1} v_{m-1} + a_m v_m + a_{m+1} v_{m+1} + \dots + a_n v_n = \mathbf{0}.$$

It can be rewritten as

$$v_m = \left(-\frac{a_1}{a_m}\right) v_1 + \left(-\frac{a_2}{a_m}\right) v_2 + \dots + \left(-\frac{a_{m-1}}{a_m}\right) v_{m-1} + \left(-\frac{a_{m+1}}{a_m}\right) v_{m+1} + \dots + (-a_n) v_n.$$

This shows that v_m is linear combination of the remaining vectors.

Conversely assume that some vector v_m of S is linear combination of the remaining vectors of S , that is, there exists some real numbers $b_1, b_2, \dots, b_{m-1}, b_{m+1}, \dots, b_n$ such that

$$v_m = b_1 v_1 + b_2 v_2 + \dots + b_{m-1} v_{m-1} + b_{m+1} v_{m+1} + \dots + b_n v_n.$$

It can be rewritten as

$$b_1 v_1 + b_2 v_2 + \dots + b_{m-1} v_{m-1} + (-1)v_m + b_{m+1} v_{m+1} + \dots + b_n v_n = \mathbf{0}.$$

We see that scalar with the vector v_m is -1 , which is non-zero. It follows that the set S is LD.

Theorem: A non-empty finite subset S of a vector space V is LI iff every vector $v \in L(S)$ can be expressed uniquely as a linear combination of the members of S .

Proof: Let $S = \{v_1, v_2, \dots, v_n\}$ be a subset of the vector space V . Suppose S is LI and v is any member of $L(S)$. Then v can be expressed as linear combination of the members of S . For uniqueness, let

$$a_1v_1 + a_2v_2 + \dots + a_nv_n = v,$$

$$b_1v_1 + b_2v_2 + \dots + b_nv_n = v.$$

Subtracting the two expressions, we get

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \dots + (a_n - b_n)v_n = \mathbf{0}.$$

Then linear independence of the set $S = \{v_1, v_2, \dots, v_n\}$ implies that $a_1 - b_1 = 0$, $a_2 - b_2 = 0$, ..., $a_n - b_n = 0$, that is, $a_1 = b_1$, $a_2 = b_2$, ..., $a_n = b_n$. This proves the uniqueness.

Conversely assume that every vector $v \in L(S)$ can be expressed uniquely as a linear combination of the members of S . To prove that $S = \{v_1, v_2, \dots, v_n\}$ is LI, let

$$a_1v_1 + a_2v_2 + \dots + a_nv_n = \mathbf{0}.$$

Also, we have

$$0v_1 + 0v_2 + \dots + 0v_n = \mathbf{0}.$$

The above two expressions represent $\mathbf{0} \in L(S)$ as linear combinations of members of the set S . So by uniqueness, we must have $a_1 = 0$, $a_2 = 0$, ..., $a_n = 0$. It follows that S is LI.

Note: An infinite subset S of a vector space V is LI iff every finite subset of S is LI. For example, the set $S = \{1, x, x^2, \dots\}$ is an infinite LI set in P (the vector space of all polynomials).

Homework: Do exercise 4.4 from the textbook.

Basis

A subset B of a vector space V is a basis of V if B is LI and $L(B) = V$. Therefore, the basis set B is LI and generates or spans the vector space V .

Ex. The set $B = \{[1, 0], [0, 1]\}$ is a basis of R^2 , called the standard basis of R^2 .

For, $B = \{[1, 0], [0, 1]\}$ is LI since $a[1, 0] + b[0, 1] = [0, 0]$ yields $a = 0$, $b = 0$.

Also, $L(B) = R^2$ since any $[x_1, x_2] \in R^2$ can be written as

$$[x_1, x_2] = x_1[1, 0] + x_2[0, 1],$$

a linear combination of the members of the set $B = \{[1, 0], [0, 1]\}$.

Ex. The set $B = \{[1, 0, 0], [0, 1, 0], [0, 0, 1]\}$ is the standard basis of R^3 .

Ex. The set $B = \{[1, 2, 1], [2, 3, 1], [-1, 2, -3]\}$ is a basis of R^3 .

Sol. Using suitable row transformations, we find

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ -1 & 2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This shows that B is LI. Also, $L(B) = \{a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) : a, b, c \in R\} = \{(a, b, c) : a, b, c \in R\} = R^3$. So B is a basis of R^3 .

Ex. The set $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is the standard basis of M_{22} .

Ex. The set $B = \{1, x, x^2, \dots, x^n\}$ is standard basis of P_n .

Sol. Any polynomial in x of degree n is, of course, a linear combination of the members of the set $B = \{1, x, x^2, \dots, x^n\}$. So $L(B) = P_n$.

Also, $B = \{1, x, x^2, \dots, x^n\}$ is LI since $a_0 \cdot 1 + a_1 x + \dots + a_n x^n = \mathbf{0} = 0 \cdot 1 + 0 \cdot x + \dots + 0 \cdot x^n$ gives $a_0 = 0, a_1 = 0, \dots, a_n = 0$.

Thus $B = \{1, x, x^2, \dots, x^n\}$ is a basis of P_n , also known as standard basis of P_n .

Ex. The empty set $\{\}$ is basis of the trivial vector space $V = \{0\}$.

Theorem: If B_1 is a finite basis of a vector space V , and B_2 is any other basis of V , then B_2 has same number of vectors as in B_1 .

Note: A vector space may have infinitely many finite bases. However, the number of vectors in each basis would be same as suggested by the above theorem. This fact led to the following definition.

Dimension

The number of elements in the basis of a vector space is called its dimension. Further, a vector space with finite dimension is called a finite dimensional vector space.

Ex. The basis $B = \{[1, 0], [0, 1]\}$ of R^2 carries two vectors. So $\dim(R^2) = 2$.

Ex. The basis set $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ of M_{22} carries four vectors. So $\dim(M_{22}) = 4$.

The following theorem gives the idea to find basis of a vector space from a spanning set.

Theorem: Any maximal LI subset of a spanning set of a vector space forms a basis of the vector space.

LI test method to find maximal LI subset

Write the given vectors as the columns in a matrix, and find its REF. The columns carrying the leading entries correspond to the LI vectors.

Ex. Find a maximal LI subset of the set $S = \{[1, 0], [2, 1], [1, 5]\}$.

Sol. Writing the given vectors as the columns, we get the matrix

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 5 \end{bmatrix}$$

Using suitable row operations, we find

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -9 \\ 0 & 1 & 5 \end{bmatrix}$$

We see that the first two columns carry the leading entries. So the corresponding set of vectors $\{[1, 0], [2, 1]\}$ is a maximal LI subset of the given set S of vectors.

Note. It should be noted that the maximal LI set given by the LI test method is not unique. For, the set $\{[1, 0], [1, 5]\}$ is also a maximal LI subset of S . So LI test method just serves the purpose of providing a maximal LI subset of the given set of vectors.

Ex. The set $S = \{[1, 0], [2, 1], [1, 5]\}$ spans R^2 . Find a basis of R^2 from this set.

Sol. As shown in the previous example that $\{[1, 0], [2, 1]\}$ is a maximal LI subset of S . So it forms a basis.

Theorem: Every LI subset of a vector space can be extended to form a basis of the vector space.

Ex. The set $S = \{[1, 3, 7]\}$ is a LI subset of R^3 . It is easy to verify that the extended set $\{[1, 3, 7], [0, 1, 0], [0, 0, 1]\}$ is a basis of R^3 .

Interesting Note: If we remove one or more vectors from the basis set of a vector space, it is no more a spanning set of the vector space. Further, if we insert one or more vectors in the basis set, it is no more LI. Thus, number of elements in the basis set does not vary.

Theorem: If W is subspace of a vector space V , then $\dim(W) \leq \dim(V)$.

Ex. Find a basis of a subspace of R^4 spanned by the set $S = \{[3, -5, 2, 4], [1, -2, 2, 1], [-1, 2, -1, 1]\}$.

Sol. We need to find maximal LI subset of $S = \{[3, -5, 2, 4], [1, -2, 2, 1], [-1, 2, -1, 1]\}$. Using suitable row operations, we find

$$\begin{bmatrix} 3 & -5 & 2 & 4 \\ 1 & -2 & 2 & 1 \\ -1 & 2 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 15 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

This shows that $S = \{[3, -5, 2, 4], [1, -2, 2, 1], [-1, 2, -1, 1]\}$ is LI. So $S = \{[3, -5, 2, 4], [1, -2, 2, 1], [-1, 2, -1, 1]\}$ itself is a basis of the subspace of R^4 .

Note: Row vectors of the row equivalent matrix of the basis vectors also form a basis. So in the above example, the set $\{[1, 0, 0, 15], [0, 1, 0, 9], [0, 0, 1, 2]\}$ is also a basis of the subspace of R^4 .

Homework: Do exercise 4.5 and 4.6 from the textbook.

Coordinatization

Let $B = (v_1, v_2, \dots, v_n)$ be an ordered basis (an ordered n -tuple of vectors) of a vector space V . Suppose $v \in V$. Then there exists real numbers $a_1, a_2, \dots, a_n$ such that $v = a_1v_1 + a_2v_2 + \dots + a_nv_n$. The n -vector $[v]_B = [a_1, a_2, \dots, a_n]$ is called the coordinatization of v with respect to B . We also say that v is expressed in B -coordinates.

Ex. The set $B = ([1, 0], [0, 1])$ is an ordered basis of R^2 . Then $[5, 4]_B = [5, 4]$ since $[5, 4] = 5[1, 0] + 4[0, 1]$.

Ex. The set $C = ([0, 1], [1, 0])$ is an ordered basis of R^2 . Then $[5, 4]_C = [4, 5]$ since $[5, 4] = 4[0, 1] + 5[1, 0]$.

Ex. Let V be a subspace of R^3 spanned by the ordered basis $B = ([2, -1, 3], [3, 2, 1])$ and $[5, -6, 11] \in V$. Then $[5, -6, 11]_B = [4, -1]$.

Sol. Let $[5, -6, 11] = a[2, -1, 3] + b[3, 2, 1]$. This yields the following system of linear equations:

$$2a + 3b = 5; \quad -a + 2b = -6; \quad 3a + b = 11.$$

Writing the augmented matrix and applying suitable row transformations, we find

$$\begin{bmatrix} 2 & 3 & \vdots & 5 \\ -1 & 2 & \vdots & -6 \\ 3 & 1 & \vdots & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \vdots & 4 \\ 0 & 1 & \vdots & -1 \\ 0 & 0 & \vdots & 0 \end{bmatrix}.$$

This gives $a = 4$ and $b = -1$. So $[5, -6, 11]_B = [4, -1]$.

Transition Matrix

Let V be a non-trivial n -dimensional vector space with ordered bases B and C . Let P be n -square matrix whose i th column is $[b_i]_C$ where $[b_i]$ is the i th basis vector in B . Then P is called the transition matrix from B -coordinates to C -coordinates or B to C . For any vector $v \in V$, it can be proved that $P[v]_B = [v]_C$.

Ex. The sets $B = ([1, 0, 0], [0, 1, 0], [0, 0, 1])$ and $C = ([1, 5, 1], [1, 6, -6], [1, 3, 14])$ are two ordered bases of

R^3 . Then $P = \begin{bmatrix} -102 & 20 & 3 \\ 67 & -13 & -2 \\ 36 & -7 & -1 \end{bmatrix}$.

$$\text{For, } \begin{bmatrix} 1 & 1 & 1 & \vdots & 1 & 0 & 0 \\ 5 & 6 & -3 & \vdots & 0 & 1 & 0 \\ 1 & -6 & 14 & \vdots & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & \vdots & -102 & 20 & 3 \\ 0 & 1 & 0 & \vdots & 67 & -13 & -2 \\ 0 & 0 & 1 & \vdots & 36 & -7 & -1 \end{bmatrix}.$$

Theorem: Suppose B , C and D are ordered bases for a non-trivial finite dimensional vector space V . Let P be transition matrix from B to C , and Q be transition matrix from C to D . Then QP is the transition matrix from B to D .

Theorem: If A is an n -square diagonalizable matrix, that is, there exists a non-singular matrix P such that $P^{-1}AP = D$, and B is an ordered basis of R^n consisting of eigen vectors of A (or column vectors of P), then for any $v \in R^n$, we have $D[v]_B = [Av]_B$.

For, if S standard basis of R^n , then P is transition matrix from B to S . So $P[v]_B = [v]_S$ and hence

$$D[v]_B = P^{-1}AP[v]_B = P^{-1}A[v]_S = P^{-1}[Av]_S = [Av]_B.$$

Homework: Do exercise 4.7 from the textbook.

Chapter 5 (5.1-5.4)

Linear Transformation

Let V and W be vector spaces. Then a function from $f : V \rightarrow W$ is said to be linear transformation (LT) iff for all $v_1, v_2 \in V$ and $c \in R$, we have $f(v_1 + v_2) = f(v_1) + f(v_2)$ and $f(cv_1) = cf(v_1)$.

Ex. $f : M_{mn} \rightarrow M_{nm}$ given by $f(A) = A^T$ is a LT.

Sol. Let $A, B \in M_{mn}$ and $a \in R$. Then we have

$$f(A + B) = (A + B)^T = A^T + B^T \text{ and } f(aA) = (aA)^T = aA^T.$$

This shows that f is a LT.

Ex. $f : R^3 \rightarrow R^3$ given by $f([x_1, x_2, x_3]) = [x_1, x_2, -x_3]$ is a LT.

Sol. Let $v_1 = [x_1, x_2, x_3], v_2 = [y_1, y_2, y_3] \in R^3$ and $a \in R$. Then we have

$$f(v_1 + v_2) = f([x_1 + y_1, x_2 + y_2, x_3 + y_3]) = [x_1 + y_1, x_2 + y_2, -(x_3 + y_3)],$$

$$f(v_1) + f(v_2) = [x_1, x_2, -x_3] + [y_1, y_2, -y_3] = [x_1 + y_1, x_2 + y_2, -(x_3 + y_3)].$$

Therefore, $f(v_1 + v_2) = f(v_1) + f(v_2)$.

$$\text{Also, } f(av_1) = f([ax_1, ax_2, ax_3]) = [ax_1, ax_2, -ax_3] = a[x_1, x_2, -x_3] = af(v_1).$$

This shows that f is a LT.

Ex. Let A be a $m \times n$ matrix. Then $f : R^n \rightarrow R^m$ given by $f(X) = AX$ is a LT.

Sol. Let $X_1, X_2 \in R^n$ and $a \in R$. Then we have

$$f(X_1 + X_2) = A(X_1 + X_2) = AX_1 + AX_2 = f(X_1) + f(X_2) \text{ and } f(aX_1) = A(aX_1) = aAX_1 = af(X_1).$$

This shows that f is a LT.

Note: A LT $f : V \rightarrow V$ is called linear operator on V .

Theorem: If $L : V \rightarrow W$ is a LT, then $L(0) = 0$ and $L(a_1v_1 + a_2v_2) = a_1L(v_1) + a_2L(v_2)$ for all $a_1, a_2 \in R$ and $v_1, v_2 \in V$.

Proof: Let $u \in V$. Then we have

$$L(0) = L(u + (-u)) = L(u) + L(-u) = L(u) + L((-1)u) = L(u) - L(u) = 0.$$

Next, $L(a_1v_1 + a_2v_2) = L(a_1v_1) + L(a_2v_2) = a_1L(v_1) + a_2L(v_2)$ since $a_1v_1, a_2v_2 \in V$ and L is a LT.

Theorem: The composition of two LT is also a LT, that is, if $L_1 : V_1 \rightarrow V_2$ and $L_2 : V_2 \rightarrow V_3$ are LT, then $L_2 \circ L_1 : V_1 \rightarrow V_3$ is a LT.

Proof: Let $u, v \in V_1$ and $a \in R$. Given that $L_1 : V_1 \rightarrow V_2$ and $L_2 : V_2 \rightarrow V_3$ are LT, we have

$$(L_2 \circ L_1)(u + v) = L_2(L_1(u + v)) = L_2(L_1(u) + L_1(v)) = L_2(L_1(u)) + L_2(L_1(v)) = (L_2 \circ L_1)(u) + (L_2 \circ L_1)(v).$$

$$(L_2 \circ L_1)(au) = L_2(L_1(au)) = L_2(aL_1(u)) = aL_2(L_1(u)) = a(L_2 \circ L_1)(u).$$

This shows that $L_2 \circ L_1$ is a LT.

Theorem: If $L : V \rightarrow W$ is a LT, and V_1 and W_1 are subspaces of V and W respectively, then $L(V_1) = \{L(v) : v \in V_1\}$ is a subspace of W and $L^{-1}(W_1) = \{v : L(v) \in W_1\}$ is a subspace of V .

Proof: To prove that $L(V_1) = \{L(v) : v \in V_1\}$ is a subspace of W , let $L(u), L(v) \in L(V_1)$ and $a \in R$.

Then $u, v \in V_1$, and we have

$$L(u) + L(v) = L(u + v) \in L(V_1) \text{ since } u + v \in V_1.$$

$$\text{Also, } aL(u) = L(au) \in L(V_1) \text{ since } au \in V_1.$$

This shows that $L(V_1)$ is a subspace of W .

Likewise, it is easy to prove that $L^{-1}(W_1) = \{v : L(v) \in W_1\}$ is a subspace of V . (please try!)

Homework: Do exercise 5.1 from the textbook.

Matrix of a Linear Transformation

Let $B = (v_1, v_2, \dots, v_n)$ and $C = (w_1, w_2, \dots, w_n)$ be ordered bases of two non-trivial vector spaces V and W respectively. If $L : V \rightarrow W$ is a LT, then there exists a unique matrix A_{BC} of order $m \times n$ known as the matrix of the LT L such that $A_{BC}[v]_B = [L[v]]_C$ for all $v \in V$. Furthermore, the i th column of $A_{BC} = [L[v_i]]_C$.

Ex. Find the matrix of the LT $L : P_3 \rightarrow R^3$ given by $L(a_3x^3 + a_2x^2 + a_1x + a_0) = [a_0 + a_1, 2a_2, a_3 - a_0]$ with respect to the ordered basis $B = (x^3, x^2, x, 1)$ for P_3 and $C = ([1, 0, 0], [0, 1, 0], [0, 0, 1])$ for R^3 . Hence compute $[L(5x^3 - x^2 + 3x + 2)]_C$.

Sol. $A_{BC} = [[L(x^3)]_C, [L(x^2)]_C, [L(x)]_C, [L(1)]_C] = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$.

Now $[5x^3 - x^2 + 3x + 2]_B = (5, -1, 3, 2)$.

$$\therefore [L(5x^3 - x^2 + 3x + 2)]_C = A_{BC}[5x^3 - x^2 + 3x + 2]_B = \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix}.$$

Theorem: Let $L : V \rightarrow W$ be a LT where V and W are non-trivial vector spaces. Let A_{BC} be matrix of L with respect to the ordered bases B and C . Suppose D and E are any other ordered bases of V and W respectively. Let P be transition matrix from B to D , and Q be the transition matrix from C to E . Then the matrix A_{DE} of L with respect to the bases D and E is $A_{DE} = QA_{BC}P^{-1}$.

Note: (i) Please do example 5 from the text book.

(ii) If $L : V \rightarrow V$ is linear operator on V , then $A_{BB} = P^{-1}A_{CC}P$, where B and C are two ordered bases of V , and P is a transition matrix from B to C .

(iii) If A_{BC} and A_{CD} are respectively the matrices of the linear transformations $L_1 : V_1 \rightarrow V_2$ and $L_2 : V_2 \rightarrow V_3$, where B, C and D are ordered bases of V_1, V_2 and V_3 respectively, then the matrix of $L_2 \circ L_1 : V_1 \rightarrow V_3$ is $A_{CD}A_{BC}$.

Homework: Do exercise 5.2 from the textbook.

Kernel and Range

Let $L : V \rightarrow W$ be a LT. Then the sets $\text{Ker}(L) = \{v \in V : L(v) = 0\}$ and $\text{Range}(L) = \{L(v) : v \in V\}$ are respectively defined as the kernel and range of L .

Ex. If $L : V \rightarrow W$ be a LT, then $\text{Ker}(L) = \{v \in V : L(v) = 0\}$ is a subspace of V and $\text{Range}(L) = \{L(v) : v \in V\}$ is a subspace of W .

Sol. We know that $0 \in V$ and $L(0) = 0$. So $0 \in \text{Ker}(L)$ and $L(0) \in \text{Range}(L)$. This shows that $\text{Ker}(L)$ and $\text{Range}(L)$ both are non-empty. To show that $\text{Ker}(L)$ is subspace of V assume that $u, v \in \text{Ker}(L)$ and $a \in R$. Then $L(u) = 0$ and $L(v) = 0$. Also, L being a LT, we have $L(u + v) = L(u) + L(v) = 0 + 0 = 0$. So $u + v \in \text{Ker}(L)$. Next, $L(au) = aL(u) = a0 = 0$. So $au \in \text{Ker}(L)$. Thus, $\text{Ker}(L)$ is a subspace of V . Likewise, it is easy to show that $\text{Range}(L) = \{L(v) : v \in V\}$ is a subspace of W . (please try!).

Ex. If $L : R^3 \rightarrow R^3$ is given by $L([x_1, x_2, x_3]) = [x_1, x_2, 0]$, then find $\text{Ker}(L)$ and $\text{Range}(L)$.

Sol. Let $[x_1, x_2, x_3] \in \text{Ker}(L)$. Then by definition of kernel, we have

$L([x_1, x_2, x_3]) = [0, 0, 0]$ or $[x_1, x_2, 0] = [0, 0, 0]$. Thus, we get $x_1 = 0, x_2 = 0$ and x_3 is any real number.

So $\text{Ker}(L) = \{[0, 0, x_3] : x_3 \in R\}$.

Given that $L([x_1, x_2, x_3]) = [x_1, x_2, 0]$. So we have $\text{Range}(L) = \{[x_1, x_2, 0] : x_1, x_2 \in R\}$.

Ex. If $L : P^3 \rightarrow P^2$ is given by $L(ax^3 + bx^2 + cx + d) = 3ax^2 + 2bx + c$, then $\text{Ker}(L) = \{d : d \in R\}$ and $\text{Range}(L) = \{3ax^2 + 2bx + c : a, b, c \in R\}$.

For, let $ax^3 + bx^2 + cx + d \in \text{Ker}(L)$. Then $L(ax^3 + bx^2 + cx + d) = 0$, or $3ax^2 + 2bx + c = 0$ gives $a = b = c = 0$ and d any real number.

Ex. Let $L : R^5 \rightarrow R^4$ be given by $L(X) = AX$, where $A = \begin{bmatrix} 8 & 4 & 16 & 32 & 0 \\ 4 & 2 & 10 & 22 & -4 \\ -2 & -1 & -5 & -11 & 7 \\ 6 & 3 & 15 & 33 & -7 \end{bmatrix}$. Find bases of $\text{Ker}(L)$ and $\text{Range}(L)$.

Sol. Here $\text{Ker}(L)$ is the solution of $AX = 0$. Also, $A \sim \begin{bmatrix} 1 & 1/2 & 0 & -2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

It follows that $x_1 = -(1/2)x_2 + 2x_4$, $x_3 = -3x_4$ and $x_5 = 0$. Let $x_2 = a$ and $x_4 = b$. Then $[x_1, x_2, x_3, x_4, x_5] = [-(1/2)a + 2b, a, -3b, b, 0] = a[-1/2, 1, 0, 0, 0] + b[2, 0, -3, 1, 0]$.

Hence $\text{Ker}(L) = \{[-1/2a + 2b, a, -3b, b, 0] : a, b \in R\}$, and $\{[-1/2, 1, 0, 0, 0], [2, 0, -3, 1, 0]\}$ is the basis of $\text{Ker}(L)$.

Range space of L is spanned by the column vectors of $A \sim \begin{bmatrix} 1 & 1/2 & 0 & -2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

We see that the first third and fifth column vectors are LI. So $\{[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0]\}$ is a basis of $\text{Range}(L)$.

Theorem: If $L : R^n \rightarrow R^m$ is a LT with matrix A with respect to any bases of R^n and R^m , then

- (i) $\dim(\text{Range}(L)) = \text{Rank}(A)$
- (ii) $\dim(\text{Ker}(L)) = n - \text{Rank}(A)$
- (iii) $\dim(\text{Ker}(L)) + \dim(\text{Range}(L)) = n$.

Verify this theorem for the previous example (important!). A generalized version of part (iii) in the above theorem is given by the dimension theorem.

Dimension Theorem: If $L : V \rightarrow W$ is a LT and V is finite dimensional, then $\dim(\text{Ker}(L)) + \dim(\text{Range}(L)) = \dim(V)$.

Homework: Do exercise 5.3 from the textbook.

One-to-One and Onto Linear Transformation

A LT $L : V \rightarrow W$ is said to be one-to-one iff $L(v_1) = L(v_2) \implies v_1 = v_2$ for all $v_1, v_2 \in V$. Further, it is said to be onto iff for any $w \in W$, there exists some vector $v \in V$ such that $L(v) = w$. So $L : V \rightarrow W$ is onto iff $\text{Range}(L) = W$.

Ex. Show that $L : R^3 \rightarrow R^3$ given by $L[x_1, x_2, x_3] = [x_1, x_2, 5x_3]$ is both one-to-one and onto.

Sol. Let $v_1 = [x_1, x_2, x_3], v_2 = [y_1, y_2, y_3] \in V = R^3$. Then

$$L(v_1) = L(v_2) \implies L([x_1, x_2, x_3]) = L([y_1, y_2, y_3]) \implies [x_1, x_2, 5x_3] = [y_1, y_2, 5y_3].$$

So $x_1 = y_1, x_2 = y_2, x_3 = y_3$ and $v_1 = v_2$. Hence, L is one-to-one.

Let $w = [z_1, z_2, z_3] \in W = R^3$. Then $v = [z_1, z_2, (1/5)z_3]$ such that $L(v) = L([z_1, z_2, (1/5)z_3]) = [z_1, z_2, z_3] = w$. This shows that L is onto.

Ex. $L : R^3 \rightarrow R^2$ given by $L[x_1, x_2, x_3] = [x_1, x_2]$ is onto but not one-to-one.

Sol. Here $V = R^3$ and $W = R^2$. Let $w = [y_1, y_2] \in R^2$. Then for any real number y_3 , we have $v = [y_1, y_2, y_3] \in R^3$ such that $L(v) = L[y_1, y_2, y_3] = [y_1, y_2] = w$. This shows that L is onto.

Now, $[1, 2, 4], [1, 2, 6] \in R^3$ such that $L[1, 2, 4] = [1, 2] = L[1, 2, 6]$, but $[1, 2, 4] \neq [1, 2, 6]$. So L is not one-to-one.

Ex. $L : R^2 \rightarrow R^3$ given by $L[x_1, x_2] = [x_1 - x_2, x_1 + x_2, x_1]$ is one-to-one but not onto.

Sol. Here $V = R^2$ and $W = R^3$. Let $v_1 = [x_1, x_2], v_2 = [x_1, x_2] \in R^2$ such that $L(v_1) = L(v_2)$ or $L[x_1, x_2] = L[y_1, y_2]$ or $[x_1 - x_2, x_1 + x_2, x_1] = [y_1 - y_2, y_1 + y_2, y_1]$. This leads to $x_1 = y_1$ and $x_2 = y_2$. So $[x_1, x_2] = [y_1, y_2]$ or $v_1 = v_2$. Thus, L is one-to-one.

Next, $[1, 2, 3] \in R^3$. Let $[x_1, x_2] \in R^2$ such that $L[x_1, x_2] = [1, 2, 3]$ or $[x_1 - x_2, x_1 + x_2, x_1] = [1, 2, 3]$. This yields the following system of equations:

$$x_1 - x_2 = 1; \quad x_1 + x_2 = 2; \quad x_1 = 3,$$

which has no solution. This in turn implies that $[1, 2, 3] \in R^3$ has no pre-image in R^2 . Hence L is not onto.

Theorem: A LT $L : V \rightarrow W$ is one-to-one iff $\text{Ker}(L) = \{0\}$.

Proof: First assume that $L : V \rightarrow W$ is one-to-one. We shall prove that $\text{Ker}(L) = \{0\}$. Let $v \in \text{Ker}(L)$. Then $L(v) = 0$, by definition of kernel. Also, we know that $L(0) = 0$. So $L(v) = L(0)$. Given that L is one-to-one. It follows that $v = 0$. So $\text{Ker}(L) = \{0\}$.

Conversely assume that $\text{Ker}(L) = \{0\}$. To prove that L is one-to-one, let $v_1, v_2 \in V$ such that $L(v_1) = L(v_2)$. So we have $L(v_1) - L(v_2) = 0$ or $L(v_1 - v_2) = 0$ since L is a LT. Further, $L(v_1 - v_2) = 0$ implies that $v_1 - v_2 \in \text{Ker}(L)$. But $\text{Ker}(L) = \{0\}$. So we get $v_1 - v_2 = 0$ or $v_1 = v_2$. Hence, L is one-to-one.

Theorem: A LT $L : V \rightarrow W$ is onto iff $\dim(\text{Range}(L)) = \dim(W)$.

Proof: By definition of onto LT, $L : V \rightarrow W$ is onto iff $\text{Range}(L) = W$. Further, we know that two vector spaces are equal iff their dimensions are equal. It follows that $L : V \rightarrow W$ is onto iff $\dim(\text{Range}(L)) = \dim(W)$.

Theorem: If $L : V \rightarrow W$ is one-to-one and $S = \{v_1, v_2, \dots, v_n\}$ is LI subset of V , then $L(S) = \{L(v_1), L(v_2), \dots, L(v_n)\}$ is LI in W .

Proof: To prove that $L(S) = \{L(v_1), L(v_2), \dots, L(v_n)\}$ is LI in W , let $a_1, a_2, \dots, a_n$ be real numbers (scalars) such that

$$a_1 L(v_1) + a_2 L(v_2) + \dots + a_n L(v_n) = 0.$$

Since L is LT and $L(0) = 0$, we have

$$L(a_1 v_1 + a_2 v_2 + \dots + a_n v_n) = L(0).$$

It is given that L is one-to-one. So we get

$$a_1v_1 + a_2v_2 + \dots + a_nv_n = 0.$$

Also, it is given that $S = \{v_1, v_2, \dots, v_n\}$ is LI subset of V . So we have $a_1 = 0, a_2 = 0, \dots, a_n = 0$. Hence $L(S) = \{L(v_1), L(v_2), \dots, L(v_n)\}$ is LI in W .

Theorem: If $L : V \rightarrow W$ is onto and S spans V , then $L(S)$ spans W .

Proof: Let $S = \{v_1, v_2, \dots, v_n\}$ spans V and $w \in W$. Since $L : V \rightarrow W$ is onto, there exists some $v \in V$ such that $L(v) = w$. Now $v \in V$ and $S = \{v_1, v_2, \dots, v_n\}$ spans V . So there exist real numbers (scalars) $a_1, a_2, \dots, a_n$ such that

$$v = a_1v_1 + a_2v_2 + \dots + a_nv_n.$$

Since L is LT, we have $w = L(v) = a_1L(v_1) + a_2L(v_2) + \dots + a_nL(v_n)$.

This shows that $L(S)$ spans W .

Homework: Do exercise 5.4 from the textbook.